

1 Physical Primitives of Mind: Modeling
2 Consciousness via Symmetries and Landauer
3 Erasure

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6 **Abstract**

7 We present a mathematical synthesis and formal Lean 4 verifi-
8 cation demonstrating how cognitive architectures can emerge as at-
9 tractor states from established physical primitives, without introduc-
10 ing new fundamental physics. Starting from the Poincaré group of
11 spacetime symmetries and finite phase space, we construct a frame-
12 work for predictive processing. Spontaneous symmetry breaking gen-
13 erates necessary topological boundaries, and finite phase space sub-
14 jects these boundaries to Landauer’s thermodynamic erasure. Agency
15 arises when systems evolve along paths of least action, minimizing en-
16 tropy production via internal structural resonance with external per-

17 turbations. Scaling this dynamic to macroscopic levels aligns with
 18 emerging paradigms in neuroscience—specifically classical electrody-
 19 namic fields (ephaptic coupling)—and Kuramoto phase synchroniza-
 20 tion, culminating in a topological fixed point: a unified “Self.” We
 21 state the claim narrowly, because the subject matter invites a wider
 22 reading than the work supports: no new physics is derived here. Every
 23 physical result the chain invokes is established—Picard–Lindelöf, Kib-
 24 ble’s classification of topological defects, Landauer’s erasure bound,
 25 Gibbs’ inequality, Still’s bound on dissipation by nonpredictive in-
 26 formation, Barbalat’s lemma, Sakaguchi’s threshold, Banach’s fixed
 27 point. What is new is that they are stated in one formal language,
 28 their consistency mechanically checked, and the points where the chain
 29 relies on stipulation rather than derivation located exactly. The de-
 30 velopment declares no axioms and every structure carrying physical
 31 content is inhabited by a witness; the synchronization step is proved
 32 from a dynamics for initial data inside an explicit basin and assumed
 33 outside it, and the identification of a global section of the probability
 34 sheaf with the unity of experience is a stipulation, marked as such
 35 throughout rather than absorbed into the derivation. The step from
 36 a coarse-grained coupling kernel to the brain’s endogenous electro-
 37 magnetic field is likewise an empirical commitment and not a deduc-
 38 tion; we separate it from the coarse-graining theorem that precedes it,
 39 weaken it to a modulatory-but-integrative role, and give four condi-
 40 tions that would falsify it. The framework is nonetheless falsifiable in

one place that is its own and no rival's: recovery from sudden awakening should follow a delayed sigmoid rather than an exponential, because the variable gating it is astrocytic volume on a timescale of seconds to minutes rather than cortical phase realignment on a timescale of milliseconds. As a corollary, we show why rigid von Neumann architectures (e.g., GPUs) are thermodynamically disadvantaged or structurally constrained from reaching these specific continuous thermodynamic attractors.

1 Introduction

The pursuit of a physical theory of consciousness typically involves searching for the biological correlates of mind or treating subjective experience as an abstract computational algorithm. This paper asks a more fundamental question: If we assume only basic physical primitives—continuous symmetries (the Poincaré group), geometry, and thermodynamics—is the emergence of a unified, perceiving entity a natural physical attractor?

We propose a framework where the structural properties of consciousness are derived as physical attractor states. Establishing a deductive chain from basic principles, we demonstrate how continuous symmetries break to form boundaries, how those boundaries dissipate free energy as non-equilibrium steady states, and how their evolution along paths of least action results in physical resonance with their environment. This eventually scales into the macroscopic electromagnetic unity proposed by emerging field theories

63 of mind. To avoid critiques of “math theatre,” we delineate between literal
 64 physical mechanisms (e.g., thermodynamic erasure via Landauer’s principle,
 65 boundaries from spontaneous symmetry breaking) and theoretical mappings,
 66 utilizing Lean 4 formalization as our mathematical anchor. We adopt a per-
 67 spective aligned with Russellian Monism: objective measurements of macro-
 68 scopic electromagnetic fields constitute the third-person “outside,” while the
 69 system’s internal physical deformation to reach structural resonance consti-
 70 tutes the first-person intrinsic experience.

71 **What is new here, and what is not**

72 A framework paper about consciousness invites a particular suspicion, and it
 73 is better answered at the start than at the end. **No new physics is derived**
 74 **here.** Every physical result the chain invokes is established, and most of
 75 it is old: Picard–Lindelöf for the well-posedness of the coupled dynamics
 76 (Coddington & Levinson, 1955); Kibble’s classification of topological defects
 77 by the homotopy of the vacuum manifold (Kibble, 1976); Landauer’s bound
 78 on the heat cost of erasure (Landauer, 1961); Gibbs’ inequality for the relative
 79 entropy (Cover & Thomas, 2006); Barbalat’s lemma for the asymptotics of a
 80 dissipating trajectory (Barbalat, 1959); the threshold $K_c = 2D$ for the noisy
 81 mean-field Kuramoto model (Sakaguchi, 1988); the bound of Still, Sivak,
 82 Bell and Crooks relating dissipation to nonpredictive information (Still et
 83 al., 2012); the identity of Kawai, Parrondo and Van den Broeck equating
 84 dissipated work with the relative entropy between forward and time-reversed

85 ensembles (Kawai et al., 2007); and Banach’s fixed-point theorem (Banach,
86 1922). A physicist reading this paper will recognize all of it and would
87 establish the point in three paragraphs if we did not. Conceding it costs
88 nothing.

89 What is new is of a different kind, and it is a smaller claim than a theory
90 of consciousness usually makes. These results have been stated in one formal
91 language, their mutual consistency has been mechanically checked, and the
92 places where the chain relies on stipulation rather than derivation have been
93 located exactly. The value of a machine-checked chain is not that any link is
94 novel; it is that no link can quietly borrow from another, that a hypothesis
95 cannot be smuggled in as a definition, and that the seams — where physics
96 stops and interpretation begins — are fixed in the source rather than nego-
97 tiable in the prose. Section 2.1 reports what the check cost: three of five
98 original axioms each proved **False**, from one shared root cause, and none of
99 the three had been caught by reading.

100 The framework is not thereby unfalsifiable, and one of its predictions is
101 its own rather than a rival’s. Section 11 argues that recovery from sudden
102 awakening should follow a *delayed sigmoid* rather than an exponential, be-
103 cause the variable gating recovery is astrocytic volume and cerebrospinal-fluid
104 clearance, on a timescale of seconds to minutes, while the cortical rhythms
105 whose realignment is usually taken to constitute the recovery move in mil-
106 liseconds. That is a claim about functional form; it is fittable against data of
107 a kind already collected; and a purely electrical account of the same recovery

108 predicts a different form.

109 2 Formalization in Lean 4

110 A core contribution of this work is the verification of our deductive chain
 111 using the Lean 4 interactive theorem prover. This approach builds on recent
 112 efforts to formalize consciousness and brain semantics, such as the ‘OnlyOne’
 113 system (Scherf, 2026) and sheaf-theoretic models of neural manifolds (Inoué,
 114 2026). It is necessary to explicitly distinguish which claims have been math-
 115 ematically derived as theorems within the proof assistant, and which remain
 116 irreducible physical postulates. Table 1 maps the primary claims of this
 117 paper to their current formalization status.

Table 1: Mapping of physical claims to Lean 4 formalization status. The development declares **no axioms**: `#print axioms` on any result above reports only `propext`, `Classical.choice` and `Quot.sound`. Earlier drafts carried five axioms, three of them inconsistent; all five became class fields, so a postulate is an obligation on each instance rather than a global assertion (Section 2.1), and two of the five have since been eliminated entirely (Derivation 5). “Theorem” still does not mean “physically established”: the postulates live in hypotheses and instance obligations, and every such class is inhabited by a witness in `Examples.lean`, so no row is vacuous.

Conceptual Claim	Lean 4 Implementation	Status
Pointwise Vacuum Minimization	<code>pointwise_vacuum_of_global_min (Phase1)</code> , potential energy defined as an integral; holds a.e., and a double-well witness proves the a.e. cannot be strengthened	Theorem
Landauer’s Principle	Derived from statistical mechanics (<code>Phase3</code>); the axiom-free reversibility core becomes a heat bound via the field <code>StatisticalMechanics.heat_eq</code>	Theorem + instance postulate
Symmetry Breaking implies Boundaries	Derived from pointwise potential minimization (<code>Phase1</code>); witnessed on a double well with a $\mathbb{Z}_2$ symmetry whose minimisation hypothesis is proved, not assumed	Theorem

Conceptual Claim	Lean 4 Implementation	Status
Inevitability of Boundaries (π_0)	The domain-wall case of the Kibble mechanism (Phase1 §3): on a connected substrate, a continuous field taking values in two different connected components of the vacuum manifold must leave the vacuum somewhere (<code>exists_notMem_of_connectedComponentIn</code>). Witnessed on the double well, whose vacuum manifold is computed to be exactly $\{-1, 1\}$ and proved disconnected (Examples §11.1); the field class is non-empty and the forced wall is located. Higher homotopy (π_1 vortices, π_2 monopoles) available and not taken	Theorem
Nonpredictive information bounds dissipation	Still, Sivak, Bell and Crooks' bound is the field <code>PredictiveDissipation.still_bound</code> (Phase3.PredictiveThermodynamics). Everything downstream is derived: mutual information is defined from Mathlib's <code>klDiv</code> ; the data processing inequality for the chain $X_t \rightarrow S_t \rightarrow S_{t+1}$ proves the nonpredictive information non-negative (<code>predictiveInfo_le_mutualInfo</code>), from which $W_{\text{diss}} \geq 0$ follows rather than being assumed; a dissipation budget bounds wasted memory, and zero dissipation forces every retained bit to be predictive. Witnessed in Examples §18 by two systems over one two-bit law, one permitted to dissipate nothing and one forced to dissipate its whole memory, with a counterexample showing the no-back-action hypothesis is load-bearing. This replaces the earlier “gradient descent on σ drives $D_{KL} \rightarrow 0$ ”, which was an invalid inference	Theorem + instance postulate
Structural Resonance KL Bound	Gibbs' inequality proved; the bound is the field <code>StructuralResonance.kl_bound</code> (Phase3.KLBound). Retained: the postulate is consistent and yields $\sigma \geq 0$. No limit may be read off it	Theorem + instance postulate
Structural Resonance via Least Action	Gradient flow of σ in coupling space (Phase8), linked to <code>entropy_production_rate</code> ; finite substrates	Theorem
Phase-locking minimizes entropy production	For a <i>reciprocal</i> coupling kernel, the phase-locked field has the least entropy production of <i>any</i> field (<code>phase_locked_minimizes_entropy_of_symm</code> , Phase8 §2a). The mean-matching condition the earlier form assumed over the comparison class is derived, not imposed: the coupling term is anti-symmetric under exchanging its two sites, so total drift is conserved across phase fields (<code>total_drift_eq_of_symm</code>). Reciprocity is necessary — a one-way coupling on two sites strictly beats its own phase-locked state (Examples §16)	Theorem

Conceptual Claim	Lean 4 Implementation	Status
Electrodynamic Discretization	Lebesgue-Bochner edge weights (Phase2) sitting on faces of the complex and partitioning the region's stress-energy; mesh refinement converges, witnessed on $[0, 1]$. The scalar read off the stress-energy tensor is a marked modelling choice: the partition holds for any non-negative choice, and two instances differing only in it have different couplings (Examples §6)	Theorem + instance postulate
Coarse-graining of a symmetric coupling	<code>mesh_refinement_convergence</code> (Phase2): the discrete coupling energy is the midpoint rule over unordered edges (<code>sum_sum_mul_of_symm</code>) and converges to the continuous integral along regular refining triangulations. It is a statement about the <i>energy functional</i> , not the dynamics; <code>ContinuousNeuralField</code> is posited, not produced, and the propagation-of-chaos gap between the finite Kuramoto system and its continuum limit is open	Theorem, narrower than the prose it supports
The field is the endogenous EM field	Not formalized and not derived. The identification of the coarse-grained kernel with cortical ephaptic coupling is the framework's central empirical commitment (Section 7). Supported as <i>modulatory and integrative</i> , not as primary: measured endogenous fields of 1–5 mV/mm shift spike timing by 1–3 ms. Four falsification conditions are stated in the text	Empirical commitment
Frustration as memory capacity	Outside the regime the theorems cover. Frustration needs competing couplings; <code>phase_locked_minimizes_potential</code> , <code>ThermodynamicCover.A_pos</code> and <code>Phase4.RotatingFrame</code> §7 all require positive coupling, the last with a uniform bound $0 < a \leq A_{ij}$. Well-posedness, Lyapunov descent and <code>velocity_sq_tendsto_zero</code> carry no sign hypothesis, so <i>that</i> the motion stops is frustration-agnostic; <i>where</i> it stops is not. The memory claim is scoped out of the chain rather than repeated	Recorded gap
Phase synchronization to Unity	Sheaf-theoretic gluing (Phase5) over a <code>ThermodynamicCover</code> , whose fields are now all local: the global section is constructed by <code>probability_glue_unique</code> and the former class field <code>sync_to_section_eq</code> is a theorem about it. Witnessed by two patches built from different mass profiles, each section constructed over its own patch, whose glued state is neither of them (Examples §14). The cover's equilibrium hypothesis is no longer assumed on every instance: <code>ThermodynamicCover.ofConvergentTrajectory</code> (Phase5.EquilibriumBridge) discharges it from <code>kuramoto_tendsto_global_minimum</code> , and Examples §17.1 is a three-patch cover whose phase field is the limit of a trajectory started <i>unsynchronized</i> . The overlap-agreement hypothesis is untouched	Theorem + instance postulate

Conceptual Claim	Lean 4 Implementation	Status
Reflexive Topology (the Self)	Banach fixed point for a contracting self-prediction map (Phase6); metric, completeness and contraction are hypotheses, all three discharged on the three-site substrate (Examples §10), where global sections are proved to be exactly the finite measures on it, the metric is the uniform distance between their densities, and the self-prediction map contracts by exactly 1/2 without being constant. The avatar's link to the field is the predicate IsRestriction-Resonance , discharged on the same witness; under it a covering family of avatars determines the global state (eq_of_avatar_eq), and without it the Self theorem is unchanged (self_of_const-Resonance)	Theorem (witnessed)
Kuramoto Lyapunov descent	Descent chained to the minimum characterization by the rotating frame (Phase4_RotatingFrame); identical frequencies	Theorem
Kuramoto dynamics run	The initial value problem is well posed on all of $\mathbb{R}$: the field is globally Lipschitz and globally bounded (kuramotoField_lipschitz , kuramotoField_norm_le), and through every state at every instant passes exactly one trajectory, defined for all time (kuramoto_cauchy_problem). Along <i>every</i> trajectory of a zero-frequency system the dynamic potential converges (dynamic_potential_tendsto) and the total dissipation $\int_0^\infty \sum_i \dot{\theta}_i^2$ is finite, equal to the potential's total drop (dissipation_integral_tendsto). The motion stops: Barbalat's lemma is proved here, Mathlib lacking it, and applied, so $\sum_i \dot{\theta}_i^2 \rightarrow 0$ along every trajectory (velocity_sq_tendsto_zero). Where it stops is proved under an initial-data condition: if the phases start within a quarter turn and the potential excess starts below $a/2$, where $a > 0$ bounds the coupling below, the trajectory converges to a phase-locked configuration that is a <i>global</i> minimiser, with $r^2 = 1$ (kuramoto_tendsto_global_minimum), by a Polyak–Łojasiewicz estimate rather than a LaSalle principle. Unconditionally this is false — splay and twisted states are equilibria — so the condition cannot be removed. Witnessed on three sites, where no closed form exists (Examples §17)	Theorem (conditional)
Continuous vs. Rigid Topology	Fixed wiring support at matched resource (Phase7_Rigidity), deriving the suboptimality Phase 7 assumed; typed continuum/discrete separation	Theorem

Conceptual Claim	Lean 4 Implementation	Status
Critical Coupling Threshold ($K_c = 2D$)	Self-consistency equation (<code>Phase8.SelfConsistency</code>): at or below K_c , threshold included, only $r = 0$ solves it; above K_c <i>exactly one</i> $r > 0$ does (<code>supercritical.fixed_point.existsUnique</code>), so the non-negative solutions are $\{0\}$ below and $\{0, r\}$ above; that r tends to 0 as $K \downarrow K_c$ and increases strictly with K (<code>coherent.branch.strictMono</code>); a solution is a density reproducing its own order parameter. Conditional on the von Mises stationary density, which is an input, and on no link to a trajectory	Theorem (conditional)

118 2.1 Soundness of the formalization

119 Formalization is only as trustworthy as the axioms it rests on, and an axiom
120 that is inconsistent silently makes every theorem downstream of it vacuous
121 while leaving the build green. We report a case of exactly this, found by
122 auditing our own development, because the failure mode is easy to reproduce
123 and we have not seen it discussed in the applied-formalization literature.

124 Earlier drafts of this work declared five axioms. Three of them were
125 individually inconsistent: each, on its own, proves **False**. All three shared
126 one root cause — an axiom that constrains a symbol it does not itself bind.
127 Concretely:

- 128 • `spontaneous_symmetry_breaking_pointwise_min` took the potential-
129 energy functional as an *implicit* argument occurring only in its hypoth-
130 esis `PotentialEnergy phi = V v0`. Instantiating that functional with
131 the constant map $\lambda_. V(v_0)$ discharges the hypothesis by `rfl` and yields
132 the conclusion for an arbitrary field. With $V(v) = v^2$, $v_0 = 0$ and $\phi \equiv 1$
133 this proves $1 = 0$.

- 134 • `landauer_heat_eq` equated the free class field `heat_dissipation` with
135 a bath-entropy difference *for every instance* of the class. An instance
136 with zero heat dissipation and a bath that gains one bit gives $0 = \log 2$.
- 137 • `kl_bound_axiom` asserted $\sigma \geq D_{KL}(P\|Q)/\Delta t$ for *all* P and Q , while σ
138 is a single real fixed by a class field. Since D_{KL} is unbounded above, a
139 point mass against a uniform distribution refutes it.

140 The remaining two axioms, `phase_invariant_periodic` and `sync_to_section_eq`,
141 had the same shape — global assertions pinning free fields of `LocalSectionSynchronization`
142 — and are refutable as soon as the probability presheaf admits two distinct
143 global sections. (Both were demoted to class fields at the time; both have
144 since been removed from the class altogether, for a separate reason given in
145 Derivation 5.)

146 The correction is uniform, and we state it as a rule we now follow: *a*
147 *physical postulate that mentions a class field must be a field of that class,*
148 *never a standalone axiom quantified over all instances.* A postulate should
149 be an obligation each model discharges, not a global claim about every model.
150 Recast this way, the development declares no axioms at all. Two of the five
151 became genuine theorems — pointwise vacuum minimization is now proved
152 from $\int f = 0 \wedge f \geq 0 \Rightarrow f = 0$ a.e., and Gibbs' inequality was already one
153 — and the other three became instance obligations.

154 This makes non-vacuity the live question, since an uninhabitable class
155 makes its theorems vacuous just as surely as an inconsistent axiom did.

156 `Examples.lean` therefore constructs witnesses: a one-bit erasure system with
 157 a genuine bath that discharges `heat_eq` and reproduces $\Delta Q = k_B T \ln 2$;
 158 a perfectly resonant system discharging `kl_bound`; and a scalar field on
 159 a one-point spacetime discharging `ActionPrinciples`. Derivation 5's glu-
 160 ing structures are witnessed too, which was until recently the largest open
 161 gap: `Examples.lean` builds a `ThermodynamicCover` over a three-site sub-
 162 strate carrying the discrete topology and its Borel σ -algebra, covered by two
 163 *overlapping* open patches whose intersection is a single shared site. The
 164 local sections are germ families of a phase-indexed finite measure $\mu_\theta =$
 165 $(1 + \cos \theta)^+ \delta_{\text{mid}}$ pushed into the sheafification of the probability presheaf; the
 166 family is provably non-constant in θ (`phaseMeasure_not_const`), so the pe-
 167 riodicity obligation of the constructor `LocalSectionSynchronization.of-`
 168 `InvariantMeasure` is discharged by an argument about `cos` rather than satis-
 169 fied trivially, and `thermodynamic_equilibrium` is *derived* from `phase_locked minimizes_potential`
 170 rather than assumed. `global_section_from_thermodynamics` then fires on
 171 this instance and glues the two patch-local sections into a unique global one.
 172 Two further covers answer the follow-up question of how non-degenerate such
 173 a witness can be made. `Examples.lean` §13 has a negative half worth record-
 174 ing: the phase field can be made non-constant as a function, but not as a
 175 physical state, because `thermodynamic_equilibrium` forces every instance
 176 to a phase-locked configuration. `Examples.lean` §14 is built the other way
 177 round, from two patches with independently chosen mass profiles that meet
 178 only on the site they share, and the section the gluing returns is neither of

179 them. That is the same kind of result as `contracting_implies_const` in
180 Derivation 6 — a proof that the witness is as strong as the class permits,
181 rather than a witness that could be improved by trying harder.

182 What the witnesses establish is that the classes are inhabitable, so the
183 theorems about them are not vacuous. What they do not establish is that
184 a cortex is such an instance. In the cover above the phase field is already
185 constant — every patch locked at $\theta = 0$ — which is precisely what makes
186 the equilibrium obligation dischargeable; getting a physical system into that
187 configuration is the content of the informal dynamical argument, not of the
188 Lean development.

189 The same audit turned up a defect of a different kind, worth record-
190 ing because it is not an axiom problem and would not have been caught
191 by `#print axioms. mesh_refinement_convergence` — the claim that dis-
192 crete edge weights converge to the continuous energy as the mesh is refined
193 — sat in a “Conjectures” section as an unproved `def ...: Prop`. It was
194 not merely unproved but false. Nothing required the edge regions to cover
195 the manifold, and $\text{diam } \emptyset = 0$, so the triangulation whose edge regions are
196 all empty satisfied the mesh hypothesis while contributing nothing to the
197 sum, forcing $|0 - \int f| < \varepsilon$ for every $\varepsilon > 0$. Its binder was dead as well: it
198 quantified over a triangulation and then resolved the vertex type by instance
199 search rather than through the bound variable, so the variable was never
200 used. `Phase2_MeshConvergence.lean` replaces it with a theorem. Conver-
201 gence is now stated along a *sequence* of triangulations of a named region

202 S , each required to be regular — edge regions measurable, without self-
 203 loops, disjoint across distinct edges, covering S , and anchored to the sim-
 204 plicial data in the sense that a vertex lies in the closure of every region
 205 of an edge at it. Closure rather than membership is forced: cells are dis-
 206 joint, so a shared vertex belongs to at most one of them. The manuscript’s
 207 discrete energy $\frac{1}{2} \sum_u \sum_v w(u, v) f(u)$, which samples only the first endpoint
 208 and is therefore not symmetric, is shown to equal exactly the midpoint rule
 209 $\sum_{u < v} w(u, v) \frac{1}{2} (f(u) + f(v))$ over unordered edges; without that identifica-
 210 tion, reading the double sum as a Riemann sum would be a category error.
 211 At fixed mesh the error is at most $\varepsilon \mu(S)$ with ε a modulus of continuity of
 212 the integrand, and convergence follows for uniformly continuous integrands
 213 as the mesh shrinks. The regularity conditions pull against each other, so
 214 **Examples.lean** §6 discharges them with the uniform partition of $[0, 1)$ into
 215 half-open cells and checks that the approximation genuinely moves — $\frac{1}{2}$ at
 216 $N = 1$ and $\frac{1}{4}$ at $N = 2$ for the tent function $x \mapsto |x - \frac{1}{2}|$ — rather than
 217 converging trivially. What is proved is convergence, not the $O(1/N^2)$ rate
 218 the accompanying simulation measures; that would need a midpoint error
 219 term with a second derivative, which we have not developed.

220 A third defect was one of bookkeeping across files, and it is the kind that
 221 survives any number of green builds because each half is individually cor-
 222 rect. The development carried *two* Kuramoto potentials. Lyapunov descent
 223 along the dynamics, $\dot{V} = -\sum_i \dot{\theta}_i^2$, was proved for the full potential $V(\theta) =$
 224 $-\frac{1}{2} \sum_{i,j} A_{ij} \cos(\theta_i - \theta_j) - \sum_i \omega_i \theta_i$, while the characterization of the minimum

225 — phase-locked in both directions — and everything downstream of it, in-
 226 cluding Derivation 5’s gluing, concerned the coupling term alone. Nothing
 227 chained them, and the gap was not cosmetic: `kuramoto_potential_unbounded_below`
 228 now proves that as soon as one $\omega_i \neq 0$ the full potential is unbounded below
 229 — walk θ out along ω and the bounded cosine sum cannot compensate — so
 230 it has no minimum for phase-locking to attain, and the sentence “the phase-
 231 locked state minimizes the Lyapunov potential” was, read literally, false of
 232 the functional the descent theorem was about. `Phase4_RotatingFrame.lean`
 233 closes this by formalizing the standard reduction $\theta_i \mapsto \theta_i - \Omega t$. For identical
 234 natural frequencies $\omega \equiv \Omega$, a trajectory maps to a trajectory of the zero-
 235 frequency system — which is exactly the system Derivation 5 already worked
 236 with — the two potentials coincide there, and the descent result transports
 237 onto the coupling potential whose minimizers are the phase-locked states.
 238 Phase differences, phase-locking and the magnitude of the order parameter
 239 are all proved invariant under the change of frame, so nothing observable
 240 is lost in the reduction. `Examples.lean` §7 runs the chain end to end on
 241 two oscillators with a common frequency, discharging every hypothesis. The
 242 restriction is real and we state it plainly: the reduction is exact only for
 243 identical frequencies. A genuine spread of natural frequencies leaves residual
 244 detunings in the reduced system, phase-locking then requires coupling above
 245 a critical K_c , and K_c is nowhere in our Lean development — it is studied
 246 numerically, as Table 1 records.

247 A fourth defect was dimensional, and it concerned the one row of Table 1

248 that is still numerical rather than proved. `Phase8_ContinuousField.lean`
 249 defines the critical coupling $K_c = 2D$ and a predicate for being above it.
 250 That predicate compared $\int_x \int_y K(x, y)$ against $2D$ — but a double integral
 251 against an unnormalized measure carries the substrate’s mass twice over,
 252 so a constant kernel K on a substrate of total mass m contributed Km^2 .
 253 The comparison therefore depended on how large the substrate was rather
 254 than on how strongly it was coupled, and any kernel whatever could be
 255 pushed above threshold by inflating m . `Examples.lean` §9 exhibits the fail-
 256 ure concretely: on a two-site substrate carrying counting measure, a sys-
 257 tem coupled at $K = 1/2$ against a threshold of $K_c = 1$ — a factor of two
 258 *below* threshold — had an unnormalized double integral of $2 > 1$ and so
 259 satisfied the old predicate. The fix is to require the substrate’s measure to
 260 be a probability measure, which is the normalization in which the mean-
 261 field derivation of $K_c = 2D$ is carried out; `mean_field_coupling_const`
 262 then gives that a constant kernel has mean-field strength exactly K , and
 263 `exhibits_phase_transition_const_iff` reduces the predicate to $K > 2D$.
 264 We flag what this does and does not do. It makes the predicate say something
 265 whose truth depends only on the physics, and §9 witnesses it on both sides of
 266 the threshold. It does not by itself prove $K_c = 2D$: the content of that value
 267 is the bifurcation of the self-consistency equation $r = I_1(Kr/D)/I_0(Kr/D)$
 268 at $K = 2D$, which is a separate question from whether the predicate is di-
 269 mensionally sound. That bifurcation is proved below, in both directions,
 270 from an assumed stationary density; deriving the density itself needs the

271 Fokker–Planck operator and is not done.

272 We draw no grand moral from this beyond the obvious one. The value
273 of a proof assistant here was not that it certified our argument, but that it
274 made a specific, checkable claim about what had been assumed — and that
275 claim turned out to be false in a way no amount of reading the prose would
276 have revealed.

277 **3 Axiom 1: Symmetries and Phase Space**

278 Our starting premise relies on fundamental geometry. In our Lean formaliza-
279 tion (`Phase1Primitives.lean`), continuous spacetime symmetries (e.g., the
280 Poincaré group) act upon field states, and the invariance of the system’s total
281 energy under that action is recorded by the class `SymmetryInvariantAction`.
282 Physically, it is such an invariance that Noether’s theorem converts into the
283 conservation of energy and momentum. The scope of the *formal* claim is nar-
284 rower and we state it plainly: the invariance is a class field, i.e. an assumption
285 a model must discharge, and no conserved quantity is ever constructed from
286 it. There is no Noether theorem in our Lean, and nothing downstream of this
287 section depends on one. The class is now inhabited — `Examples.lean` §11
288 exhibits a double-well potential $V(v) = (v^2 - 1)^2$ on a two-point spacetime
289 with the $\mathbb{Z}_2$ action $v \mapsto \pm v$, discharges the invariance because $(\pm v)^2 = v^2$,
290 and proves the vacuum manifold $\{-1, 1\}$ degenerate with the two minima ex-
291 changed by the symmetry, so an energy-minimising field and its image have

292 equal energy and differ. That is symmetry breaking exhibited rather than
 293 assumed; it is still not a conserved quantity, and $\mathbb{Z}_2$ being discrete there is
 294 no one-parameter family to differentiate along in the first place. Simultane-
 295 ously, discrete permutation symmetries dictate the statistical mechanics of
 296 any physical system, defining a strictly finite volume of phase space.¹

297 Because phase space is finite, any localized physical system possesses a
 298 mathematical limit on the amount of information (distinguishable states) it
 299 can embody at any given moment.

300 4 Derivation 1: Symmetry Breaking and the 301 Inevitability of Boundaries

302 A perfectly symmetric universe is the uniform vacuum; no distinct entities
 303 exist. For any dynamic process to occur, the continuous symmetries of the
 304 background field must undergo spontaneous symmetry breaking.

305 When a field drops to a lower, asymmetric energy state, topology dic-
 306 tates the natural creation of defects (such as domain walls or monopoles).

¹While the Poincaré group and discrete permutations serve as accessible physical examples, the underlying Lean formalization does not hardcode specific physics. It defines the system at a more primitive mathematical level, relying entirely on generic Pseudo-Riemannian manifolds (for spacetime) and finite measure theory (for statistical mechanics). What the formalization supplies is the vocabulary in which symmetries, conserved quantities and phase-space constraints can be *stated*; it does not derive them. All three of the classes involved are now inhabited in `Examples.lean` — the $\mathbb{Z}_2$ action and its invariance in §11, the real line with the Euclidean metric in §12 — but the instances are witnesses of inhabitability, not of physics: no theorem in the development consumes a symmetry group or a metric, so this section still contributes vocabulary rather than content.

307 This constraint necessitates the creation of a physical boundary—a localized
 308 topological distinction between an “inside” and an “outside.” While such
 309 a defect (e.g., a crack in a crystal lattice) is necessary to define a separate
 310 entity, it is not sufficient for agency. The boundary is relational, defined by
 311 the twisted geometry of the field surrounding it, and must be coupled with
 312 non-equilibrium thermodynamics to exhibit active behaviors.

313 That first sentence was, until recently, the one claim of this section that
 314 the formalization did not make. What Lean established was the energy half
 315 — an energy-minimizing field sits in the vacuum manifold almost everywhere
 316 (`pointwise_vacuum_of_global_min`) — together with the observation that
 317 a boundary defect *obstructs* extension over a contractible interior. Neither
 318 produces a defect, so “topology dictates the natural creation” was asserted
 319 rather than proved. It is now the theorem `exists_notMem_of_connected-`
 320 `ComponentIn`: if the substrate is connected and a continuous field takes val-
 321 ues in two *different connected components* of the vacuum manifold M , then
 322 somewhere the field is not in M at all. That excluded region is the wall.
 323 This is the π_0 case of Kibble’s classification of topological defects by the ho-
 324 motopy of the vacuum manifold (Kibble, 1976), and the proof is elementary
 325 in exactly the way it is elementary in the physics literature: the continuous
 326 image of a connected space is connected, a connected subset of M lies in-
 327 side one component of M , and the field’s range would be such a subset if it
 328 never left M . No hypothesis mentions energy, a potential, or a symmetry;
 329 connectedness of the substrate and disconnectedness of the vacuum are the

330 whole of it.

331 We formalize the π_0 case and stop there. The higher cases — $\pi_1 \neq 0$
332 forcing vortex lines, $\pi_2 \neq 0$ forcing monopoles — are the rest of Kibble’s
333 classification and Mathlib’s homotopy-group library would support at least
334 the next one; they are available and not taken, because the argument this
335 paper runs uses domain walls and nothing else. The witness is the double
336 well of `Examples.lean` §11.1, where the vacuum manifold is *computed* to be
337 exactly $\{-1, 1\}$ (`wellVacuum_eq`) rather than merely known to contain both
338 minima, and its disconnection is proved in the form the theorem consumes:
339 no preconnected subset of it contains both, since a preconnected subset of $\mathbb{R}$
340 is an interval and an interval spanning -1 and 1 contains 0 , which is the top
341 of the barrier. `wellV_domain_wall` then quantifies over *every* continuous field
342 on *any* connected substrate that reaches both minima, with no formula for
343 the field anywhere in the statement. That the class is non-empty is checked
344 separately, because a theorem quantified over all fields is worthless if none
345 satisfies its hypotheses: the clipped identity on $\mathbb{R}$ does, and the wall it is
346 forced to have sits at the origin (`kink_zero_notMem`).

347 5 Derivation 2: The Thermodynamic Con- 348 straint

349 This newly formed topological boundary is subjected to gauge perturbations
350 from the “outside” environment. Because the internal phase space is finite

351 (Axiom 1), it cannot absorb an infinite sequence of incoming perturbations
352 and must overwrite old internal states.

353 By Landauer’s Principle (Landauer, 1961), the erasure of physical in-
354 formation irreversibly decreases internal entropy, mandating that the sys-
355 tem exhaust heat into the environment. Philosophers of physics (such as
356 John Norton (Norton, 2005) and Orly Shenker (Shenker, 2000)) have cri-
357 tiqued Landauer’s principle, arguing that physicists conflate *logical* erasure
358 (a many-to-one mapping in abstract computational states) with *thermody-*
359 *namic* erasure (a many-to-one mapping in physical phase space). They argue
360 that logical erasure could conceptually be implemented by an injective phys-
361 ical process without dissipating heat.

362 Our Lean 4 formalization (`Phase3.CombinatorialThermodynamics.lean`)
363 addresses this critique by defining erasure as a non-injective mapping on the
364 *physical* state space ($\neg \text{Function.Injective } t$). We formalize the global evolu-
365 tion of the bipartite universe (System + Bath) as strictly injective, repre-
366 senting Liouville’s theorem. Lean verifies that heat dissipation is not tied
367 to abstract logical erasure, but is a mandatory mathematical consequence of
368 any local physical state space contraction under globally reversible dynamics.
369 If a cognitive system’s internal topology forces a contraction of its physical
370 state space to accommodate structural resonance, heat dissipation becomes a
371 thermodynamic requirement. A stable boundary thus constitutes a dissipa-
372 tive structure (Prigogine & Stengers, 1984), operating as a non-equilibrium
373 steady state that consumes free energy to pay the thermodynamic cost of

374 erasure. Continuous thermal fluctuations play a constructive role in estab-
 375 lishing boundaries between states while contributing to the system’s flexible,
 376 adaptive response.

377 6 Derivation 3: Prediction as the Thermody- 378 namic Cost of Memory

379 A dissipative structure is driven by its environment. Write X_t for the state
 380 of the structure and S_t for the environmental signal driving it, and let the
 381 signal evolve by a dynamics of its own, $S_t \rightarrow S_{t+1}$, on which the structure
 382 does not act. Two quantities are then defined at each step: the *memory*
 383 $I(X_t; S_t)$, the mutual information between the structure’s present state and
 384 the signal’s present value, and the *predictive information* $I(X_t; S_{t+1})$, which
 385 is what that same state says about the signal’s next value.

386 **What this section used to claim, and why it is withdrawn.** Earlier
 387 versions of this derivation argued as follows. Model the environment as a
 388 stochastic driving force $P(ext)$ and the structure’s internal transitions as
 389 $Q(int)$; entropy production σ is then bounded below by $D_{KL}(P||Q)/\Delta t$;
 390 the principle of least action drives σ downward; therefore $D_{KL}(P||Q) \rightarrow$
 391 0, and the internal transition probabilities come to mirror the statistical
 392 symmetries of the environment. The final step does not follow. Rearranged,
 393 the bound reads $D_{KL}(P||Q) \leq \Delta t \sigma$, so descending on σ lowers a *bound* on

the divergence; but a driven system does not relax to $\sigma = 0$. It relaxes to a non-equilibrium steady state at strictly positive entropy production, and the bound then yields only $D_{KL} \leq \Delta t \sigma_{\text{NESS}}$, a positive number. Nothing forces the divergence to zero. The inequality itself is true, is formalized as an instance obligation, and does real work — combined with Gibbs’ inequality it gives $\sigma \geq 0$ — but the conclusion drawn from it was not licensed, and it was that conclusion which supplied this framework’s entire connection to predictive processing.

The replacement is a published theorem. Still, Sivak, Bell and Crooks (Still et al., 2012) prove that for a system driven by an environmental signal the work dissipated over one step of the drive obeys

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (1)$$

The bracket is the *nonpredictive* information: what the structure retains about the signal’s present that says nothing about the signal’s future. Dissipation is bounded below by it. Memory that fails to predict is not merely useless to the structure; it is thermodynamic waste, and the waste is quantified in energy.

Three consequences, and they are theorems rather than readings. The bound is carried in Lean as a class field, `PredictiveDissipation`. `still_bound (Phase3.PredictiveThermodynamics)`, for the same reason ev-

413 ery other physical postulate here is: its `axiom` form, quantified over all joint
 414 laws with the dissipated work fixed, is refutable, and `Examples §18.4` refutes
 415 it. What is *derived* is everything downstream.

416 First, *the bracket is non-negative*, and this is proved rather than assumed.
 417 If the structure does not act on its environment, then $X_t \rightarrow S_t \rightarrow S_{t+1}$
 418 is a Markov chain and the data processing inequality gives $I(X_t; S_{t+1}) \leq$
 419 $I(X_t; S_t)$: a system cannot know more about the signal’s future than about its
 420 present (`predictiveInfo_le_mutualInfo`). Mathlib carries the Kullback–
 421 Leibler divergence but no mutual information, so the latter is defined here
 422 as the divergence of the joint law from the product of its marginals, which is
 423 the standard definition and makes the library’s data processing lemma apply
 424 directly.

425 Second, *the second law follows* instead of being assumed. Dissipation is
 426 bounded below by a quantity now known to be non-negative, so $W_{\text{diss}} \geq 0$
 427 (`dissipatedWork_nonneg`). This mirrors the role Gibbs’ inequality plays for
 428 the older bound, with one difference that matters: Gibbs’ inequality com-
 429 pares a system to an arbitrary reference distribution, whereas the data pro-
 430 cessing inequality compares the system to its own future, which is a quantity
 431 selection can act on.

432 Third, and this is the point of the replacement, *the inference now runs*
 433 *in the valid direction*. The quantity of interest sits on the small side of
 434 the inequality: $I(X_t; S_t) - I(X_t; S_{t+1}) \leq W_{\text{diss}}/k_B T$. No limit is required.
 435 Any bound on dissipation is immediately a bound on wasted memory, so a

436 structure held to an energy budget is thereby held to a prediction standard:
 437 this is `nonpredictive_le_dissipation`. In the idealised quasi-static case
 438 $W_{\text{diss}} = 0$, the theorem `predictive_eq_of_no_dissipation` gives that every
 439 retained bit is predictive. This is a weaker statement than the one it replaces,
 440 and it is the one that is true.

441 **The hypothesis is load-bearing, and we exhibit what breaks with-**
 442 **out it.** The data processing step requires that the structure not write to
 443 its environment. In the formalization this is not an assumption stated beside
 444 the theorem but the shape of the evolution itself: the signal's step is applied
 445 to the pair as the parallel kernel $\text{id} \parallel \kappa$, which acts as the identity on the
 446 system coordinate and through κ on the signal coordinate, so no channel
 447 from X_t to S_{t+1} exists. **Examples** §18.3 shows the restriction is not decora-
 448 tion: a Markov kernel on the *pair* which copies the system's state into the
 449 signal carries an independent two-bit law to a perfectly correlated one, rais-
 450 ing mutual information from zero to a positive value. An agent that steers
 451 its environment can indeed know more about that environment's future than
 452 about its present, and Still's bound is not about such an agent.

453 **Both regimes are exhibited, not just described.** **Examples** §18 builds
 454 two systems over one correlated two-bit joint law, differing only in the signal's
 455 dynamics. When the signal is frozen, every remembered bit is a predictive
 456 bit, the nonpredictive information is zero, and the instance is permitted to
 457 dissipate nothing — with the memory itself proved strictly positive, so the

458 conclusion is about a system that does remember something. When the signal
 459 is redrawn from its stationary law at each step, the predictive information is
 460 exactly zero, the *entire* memory is nonpredictive, and the instance is *forced*
 461 to dissipate $k_B T$ times its whole mutual information. Establishing that the
 462 mutual information of the witness is neither zero nor infinite is where the
 463 work of that section lies.

464 **An equality, where the old argument had only a bound.** Kawai,
 465 Parrondo and Van den Broeck (Kawai et al., 2007) show that the average
 466 dissipated work is exactly $k_B T$ times the relative entropy between the for-
 467 ward and time-reversed phase-space densities. Carried here as a predicate on
 468 an instance rather than as a further postulate, it replaces the abstract dis-
 469 sipation field with a quantity an experiment can estimate — the statistical
 470 distinguishability of a recording of the system from the same recording run
 471 backwards — and yields the statement that the irreversibility of the dynamics
 472 is at least the amount of memory that fails to earn its keep (`nonpredictive_`
 473 `le_arrow`).

474 **What this does not establish.** It does not deliver the Free Energy Prin-
 475 ciple. Still’s bound says that prediction is thermodynamically favoured; it
 476 says nothing about hierarchical generative models, variational free energy,
 477 or Bayesian inference, and we do not claim to have derived any of them.
 478 The Free Energy Principle (Friston, 2010; Friston & Kiebel, 2009; Hohwy,
 479 2013; Clark, 2016; Spisák & Friston, 2026) and the General Resonance The-

480 ory of Consciousness (Hunt & Schooler, 2019) are consonant with what is
481 proved here, which is a weaker relation than derivation and the only one
482 we assert. Nor does it solve the symbol grounding problem (Harnad, 1990).
483 Maximal predictive information is compatible with the structure storing a
484 lossy, compressed or otherwise unrecognisable function of the signal; a state
485 that predicts is not thereby a state that means, and earlier versions of this
486 section overstated the case. What survives of that argument is narrower and
487 still worth saying: whatever internal states a physical structure holds, those
488 that do not help predict its environment are paid for in heat, so the pressure
489 toward states indexed to the environment is energetic rather than semantic.
490 We continue to call the phenomenon *structural resonance*, and there is no
491 teleological drive to “survive” in it: predicting the environment is simply the
492 cheaper of the two ways to hold a memory.

493 **7 Macroscopic Scaling: A Theorem and a** 494 **Commitment**

495 This is the chain’s fourth step, and it is deliberately not numbered as a deriva-
496 tion. The change is not cosmetic. Earlier drafts ran two steps of different
497 kinds together in a single paragraph: a coarse-graining argument, which is
498 a piece of mathematics, and the identification of the coarse-grained medium
499 with the brain’s endogenous electromagnetic field, which is an empirical claim
500 about cortex. Presenting the second as though it inherited the licence of the

501 first made it look like a deduction, and it is not one. Separated, the empirical
 502 claim becomes the framework’s central commitment and the place where it
 503 is most easily wrong — which is where a claim of this kind belongs.

504 **The formal half: coarse-graining is a theorem, and a**
 505 **narrow one**

506 A network of discretely coupled units has a coupling energy that is a sym-
 507 metric double sum over pairs. `Phase2.MeshConvergence.lean` proves that
 508 this sum converges to a continuous integral: along a sequence of regular
 509 triangulations of a region S whose edge regions shrink to points, the dis-
 510 crete energy $\frac{1}{2} \sum_u \sum_v \mu(R_{uv}) f(x_u)$ converges to $\int_S f d\mu$ (`mesh_refinement_`
 511 `convergence`), witnessed on the uniform partition of $[0, 1)$ in Examples §6.
 512 The identification that makes this a coarse-graining rather than an arbitrary
 513 Riemann sum is `sum_sum_mul_of_symm`: because the summand is symmetric,
 514 the double sum is exactly the midpoint rule over unordered edges, so the
 515 discrete energy is a quadrature of the continuous one rather than a different
 516 quantity that happens to converge to it.

517 What this establishes is worth stating exactly, because it is less than the
 518 phrase “coarse-graining, governed by the Renormalization Group” suggests.
 519 It is a statement about an *energy functional*, not about a dynamics: noth-
 520 ing here shows that trajectories of the discrete Kuramoto system converge
 521 to solutions of a field equation, and the earlier appeal to the Renormaliza-

tion Group was decorative — no renormalization group flow is constructed anywhere in this development. The limit object is a scalar density on the substrate, in the intended reading the magnitude of the stress-energy tensor, and it is not an electromagnetic field. The continuous object the later sections use, `ContinuousNeuralField`, is *posited* as a structure with a coupling kernel K ; no theorem produces it from the discrete system. The gap between the finite Kuramoto model and its continuum limit is the propagation-of-chaos problem, which we have not addressed and record as open.

The empirical half: which physical field realizes the kernel

The formal half says that a large network of symmetrically coupled units admits a continuum description with a symmetric kernel. It says nothing about which physical field in cortex that kernel describes. Our answer is an empirical commitment, and we state it as one:

In the mammalian cortex, the field realizing the coarse-grained coupling kernel is the endogenous electromagnetic field.

This is a claim about biology, not a consequence of anything above it. It is what makes the framework falsifiable and it is what we expect to be attacked.

541 **What supports it.** Neurons do not only signal through synapses: trans-
542 membrane currents generate extracellular fields that shift the membrane po-
543 tentials of neighbouring cells, a mechanism known as ephaptic coupling. En-
544 dogenous fields of 1–5 mV/mm have been shown *in vitro* to entrain spike
545 timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015); realistic
546 head models put mesoscopic ephaptic effects within the range where they
547 can bias network dynamics (Ruffini & Santarnecchi, 2020); cable-theoretic
548 treatments show field coupling is sufficient for coincidence detection in the
549 auditory brainstem (Goldwyn & Rinzel, 2016); and stable electric fields have
550 been argued to carry information that survives representational drift in the
551 underlying spiking population (Pinotsis & Miller, 2022, 2023a,b; Cunha &
552 Lima, 2024). The field is spatially continuous and superposes contributions
553 from large populations, which is the property the framework needs: it is a
554 medium in which spatially separated activity is combined without a further
555 mechanism for combining it. This is the integrative role the CEMI field
556 theory assigns it (McFadden, 2002, 2020; Pockett, 2012).

557 **What the evidence does not support, and a claim we withdraw.**
558 Earlier drafts asserted that *the continuous macroscopic EM field is the pri-*
559 *mary dissipative structure interacting with the universe.* The word “primary”
560 is not supported by any of the work cited above and we withdraw it. What
561 the experiments show is *modulation*: Anastassiou and Koch report that en-
562 dogenous fields of 1–5 mV/mm shift spike timing by one to three millisec-

563 onds (Anastassiou & Koch, 2015), an effect that is real, reproducible, and
564 subthreshold. It biases when a cell fires; it does not decide whether it fires.
565 The corrected claim is that the endogenous field is *modulatory in its effect on*
566 *individual neurons and integrative in its spatial reach* — weak per synapse,
567 but the only variable in the system that is defined everywhere at once. The
568 framework needs the second property, not the first, and conflating them was
569 the error.

570 **Its critics.** The objection with the longest history is that the fields are
571 too weak to matter, and it has been pressed from within the field-theoretic
572 camp: Pockett, who has herself defended an electromagnetic theory of con-
573 sciousness (Pockett, 2012), catalogues its difficulties (Pockett, 2002) — the
574 measurement problem of relating gross field structure to reported experience,
575 the absence of a constant relationship between the two, and the question of
576 whether the field does causal work or merely reflects it. The quantitative case
577 for scepticism has strengthened since. Vöröslakos and colleagues measured
578 intracranial fields during transcranial stimulation in rats and human cadav-
579 ers and found that the great majority of applied current is shunted through
580 the scalp, so that conventional protocols deliver fields at or below the 1 V/m
581 range in cortex (Vöröslakos et al., 2018). That is a bound on *applied* fields
582 rather than endogenous ones, but it constrains how much causal work a field
583 of that magnitude can be doing, and it is the reason we do not claim the
584 field drives spiking.

585 **What would falsify it.** We state the failure conditions rather than leaving
586 them to a referee.

587 1. *Epiphenomenality.* If the endogenous field can be cancelled or shunted
588 — by a conductive medium that nulls the extracellular potential with-
589 out altering transmembrane currents — and neither behaviour nor re-
590 port changes, the identification fails. This is the decisive experiment
591 and it is, in principle, available.

592 2. *A magnitude gap.* If the spatial integration the framework requires
593 demands field effects an order of magnitude above the 1–5 mV/mm
594 measured in cortex, the commitment fails on arithmetic. The relevant
595 number is not whether ephaptic coupling exists but whether its coupling
596 strength suffices to place the system above the critical coupling K_c of
597 Derivation 7.

598 3. *A timescale mismatch in the wrong direction.* The prediction of Sec-
599 tion 11 turns on the field’s dynamics being fast relative to the astrocytic
600 and clearance variables. If recovery from sleep inertia tracks an expo-
601 nential set by the electrical timescale rather than the delayed sigmoid
602 the slow variable predicts, the framework loses the one prediction that
603 is its own.

604 4. *Dissociation of field coherence from reported unity.* If large-scale field
605 coherence is preserved across states in which experiential unity is lost

606 — or absent in states where it is retained — the identification of the
607 global section with unified experience has no physical carrier.

608 A framework that survives all four is not thereby true. It is merely still
609 standing, which is more than most theories of consciousness arrange to be
610 able to lose.

611 **Geometric frustration, and the regime our theorems ac-** 612 **tually cover**

613 Earlier drafts grounded the system’s memory capacity in *geometric frustra-*
614 *tion*: a highly coupled network cannot bring all its nodes to their lowest en-
615 ergy simultaneously, producing a jagged landscape of near-degenerate states
616 with the capacity to store many patterns. That argument is standard and it
617 is not what this development formalizes. Confronting the discrepancy is the
618 honest course, so we do it here rather than in a footnote.

619 Frustration requires competing couplings — in the Kuramoto setting, a
620 sign structure that no phase assignment can satisfy on every edge at once.
621 The formal setting admits this: `KuramotoSystem` carries $A : V \rightarrow V \rightarrow$
622 $\mathbb{R}$, symmetric and otherwise unconstrained, so signed couplings are repre-
623 sentable. But every theorem in this development that says anything about
624 *where* the dynamics ends requires strictly positive coupling. `phase_locked_`
625 `minimizes_potential` assumes $A_{ij} > 0$; `potential_min_iff_phase_locked`
626 inherits it; the class field `ThermodynamicCover.A_pos` demands it of every

627 instance, and hence of every conclusion of Derivation 5; and the convergence
 628 theorem of `Phase4.RotatingFrame` §7 requires the strictly stronger uniform
 629 bound $0 < a \leq A_{ij}$. A network with all-positive couplings is an unfrustrated
 630 ferromagnet whose potential has one minimum up to a global phase shift. It
 631 is the opposite of a jagged landscape, and it has no capacity to store multiple
 632 patterns at all. **The system in which memory is grounded and the**
 633 **system our theorems describe are not the same system.**

634 The division is not arbitrary, and it is worth recording where it falls. What
 635 survives with no assumption on the sign of the coupling is the dynamical half:
 636 well-posedness on all of $\mathbb{R}$ (`is_kuramoto_trajectory_exists`, `_unique`), co-
 637 variance under passage to a rotating frame, monotone descent of the potential
 638 (`dynamic_potential_antitone`), finiteness of the dissipation integral, and —
 639 through Barbalat’s lemma — that the motion stops along every trajectory
 640 (`velocity_sq_tendsto_zero`). *That* the motion stops is frustration-agnostic.
 641 *Where* it stops is not, and every claim this framework makes about the des-
 642 tination lives on the positive side of that line. On the continuum side the
 643 accounting differs slightly: `phase_locked_minimizes_entropy_of_symm` as-
 644 sumes only that the kernel is symmetric and so tolerates signed kernels, but
 645 it is conditional on a dynamically phase-locked field existing — which is
 646 precisely what frustration may prevent.

647 We therefore scope the claim rather than repeat it. The memory capacity
 648 of a frustrated cortical network is not supported by any theorem here, and we
 649 do not assert it as part of the chain. What the chain uses is the unfrustrated

650 regime, in which synchronization is reached and the global section exists.
651 Whether a frustrated network can hold many near-degenerate patterns *and*
652 still synchronize enough of itself to admit a global section is an open question
653 about mixed-sign oscillator networks, and it is the most substantive gap we
654 know of in this framework. Recording it as such costs a claim we would have
655 liked to make and is preferable to holding two incompatible pictures at once.

656 **The supporting physiology**

657 The continuous medium is physically supported by astrocytic syncytia—non-
658 neuronal networks joined by gap junctions that provide macroscopic cytoplas-
659 mic continuity for energy and ion flows (Verkhatsky & Nedergaard, 2018).
660 Managing the thermodynamic cost of this state requires sophisticated micro-
661 bioenergetics. Dynamic mitochondrial networks function as deep-level Mi-
662 tochondrial Information Processing Systems (MIPS), transducing biological
663 and environmental information to sustain the coherence of the macroscopic
664 field (Picard & Shirihaï, 2022). Discrete synaptic firings, coupled with mito-
665 chondrial micro-mechanics, generate and sustain the field. Because the exter-
666 nal environment is continuous, achieving maximal thermodynamic structural
667 resonance is optimally realized within this continuous macroscopic medium
668 (van Bree et al., 2025).

669 8 Derivation 5: Global Sections and the Unity 670 of Consciousness

671 The system is now characterized by a continuous macroscopic electrodynamic
672 field generated by millions of coupled micro-components, forced to resonate
673 with the environment. Following Section 7, the coupling regime assumed
674 from here on is the unfrustrated one — all couplings positive — which is
675 what every theorem below requires and what the frustrated network of the
676 memory argument is not.

677 The Kuramoto model (Kuramoto, 1984) of coupled non-linear oscilla-
678 tors guarantees that, past a critical coupling threshold, these distributed
679 local fields undergo a phase transition into unified synchronization (Bauer et
680 al., 2022). Our Lean proofs (`Phase4_KuramotoDynamics.lean`) confirm this
681 phase-locked state minimizes the Lyapunov potential of the system, in the
682 precise sense discussed in Section 2.1: the functional minimized is the cou-
683 pling potential read in the frame rotating with the natural frequency, which
684 `Phase4_RotatingFrame.lean` shows is the honest reduced description of a
685 system of identical oscillators.

686 This phase transition mathematically glues local synchronized states into
687 a singular *Global Section* of a probability sheaf (`Phase5_GlobalSection.lean`).
688 The EM field acts as a single, indivisible topological space, providing the
689 *Unity of Consciousness*—a necessary condition for integrated experience that
690 solves the binding problem. While Global Workspace Theory (Baars, 1997;

Dehaene & Marti, 2014) models this unity as a broadcast architecture and Integrated Information Theory (Tononi & Koch, 2016) quantifies it via Φ , our derivation provides a continuous physical mechanism. The contrast is worth drawing carefully, because both theories are addressing the same explanandum from different directions.

We state the scope of the Lean result precisely, because it is narrower than the prose above. `ThermodynamicCover` demands that the cover’s phase field minimize the reduced Kuramoto potential, and `potential_min_iff_phase_locked` characterizes those minimizers exactly: they are the phase-locked configurations, in both directions. Every instance is therefore already phase-locked (`ThermodynamicCover.phase_locked`), and the class’s single condition on its local data — that patches at a common phase agree where they overlap (`section_agrees_of_phase_eq`) — makes the family of local sections compatible. `probability_glue_unique`, which is the sheaf property of the probability presheaf and contains no physics at all, then glues that family into exactly one global section.

An earlier version of this structure proved less than the prose claimed, and we record the correction because it changed what the theorem says rather than how it is stated. `LocalSectionSynchronization` used to carry a *global* section as a class field, together with a second field declaring every local section to be a restriction of it. The global object therefore existed before the sheaf condition was ever consulted, and `global_section_from_thermodynamics` could do no more than rule out competitors to an object

714 the instance had supplied. Both fields are gone. Every field of the class
 715 now mentions a single patch or a single overlap; the global section is con-
 716 structed (`ThermodynamicCover.invariantMeasure`); and the removed field
 717 survives as a *theorem* about that construction (`ThermodynamicCover.sync_
 718 to_section_eq`), so the local sections are restrictions of one common measure
 719 because the sheaf condition says so, not because an instance was required to
 720 name one.

721 `Examples.lean` §14 is the witness that the difference is not notational.
 722 Two patches carry two different mass profiles, $(2, 1, 7)$ and $(5, 1, 3)$ across
 723 the three sites, agreeing only at the site they share, where both put mass 1
 724 — which is what discharges the overlap condition, by computing two num-
 725 bers rather than by construction. The section the theorem returns has pro-
 726 file $(2, 1, 3)$: it is neither of the two profiles (`gluedW_ne_leftW`, `gluedW_ne_
 727 rightW`), and neither of them restricts correctly to both patches (`leftW_
 728 glues_nothing`, `rightW_glues_nothing`). The global state carries informa-
 729 tion that no single patch had. `Examples.lean` §13, built the older way
 730 through the constructor `LocalSectionSynchronization.ofInvariantMeasure`,
 731 is retained alongside it for the negative result it carries: its phase field is not
 732 the constant function — the second patch a full turn ahead of the first — and
 733 by `ThermodynamicCover.phase_locked` that is as far from constant as any
 734 instance can get, since no instance can carry phases differing by anything a
 735 measure is allowed to see.

736 Two hypotheses carry physical content, and until recently each was an

737 obligation on every instance. One still is: that synchronized patches agree
 738 on their overlaps is a modelling assumption, not a consequence of any dy-
 739 namics we formalize. The other, `thermodynamic_equilibrium`, asks that the
 740 cover already sit at the potential minimum, and no theorem says an *arbi-*
 741 *trary* system’s dynamics takes it there. “A thermodynamic phase transition
 742 into unity” is therefore still doing causal work that no general Lean result
 743 supports, though it is no longer entirely without one: `Examples.lean` §15
 744 exhibits a two-oscillator trajectory that starts a quarter turn out of phase
 745 and converges to the global minimum of the dynamic potential, with the or-
 746 der parameter tending to 1. That trajectory is not an isolated construction:
 747 `kuramoto_cauchy_problem` proves the Kuramoto initial value problem well
 748 posed on all of $\mathbb{R}$, so through every initial state there passes exactly one tra-
 749 jectory, and the exhibited one is the only solution through its own. Nor is its
 750 settling down peculiar to it: along every trajectory of a zero-frequency sys-
 751 tem the dynamic potential converges, and the total dissipation $\int_0^\infty \sum_i \dot{\theta}_i^2$ is
 752 finite and equal to the potential’s total drop (`dynamic_potential_tendsto`,
 753 `dissipation_integral_tendsto`). The motion also stops: a finite integral
 754 does not by itself force its integrand to zero, but the dissipation rate is
 755 Lipschitz in time, so Barbalat’s lemma applies, and every trajectory satis-
 756 fies $\sum_i \dot{\theta}_i^2 \rightarrow 0$ (`velocity_sq_tendsto_zero`; Mathlib has no Barbalat, so it
 757 is proved here as `tendsto_zero_of_lipschitz_of_integral_le`). That last
 758 step — locating where the motion stops — is now taken, though condition-
 759 ally, and by a different route than we had recorded. We had marked it as

760 blocked on the LaSalle invariance principle, which Mathlib does not have,
 761 and on the absence of a compact invariant set in the state space $V \rightarrow \mathbb{R}$.
 762 That was the wrong diagnosis. LaSalle is the tool for a system with a de-
 763 creasing functional and nothing else; this one has more. The flow is a gra-
 764 dient flow of the dynamic potential, so $\dot{V} = -\sum_i \dot{\theta}_i^2$ is an identity rather
 765 than an inequality; the minimisers of V are characterised exactly; and on
 766 the region where every pair of phases lies within a quarter turn a Polyak-
 767 Łojasiewicz estimate holds, $2a^2W \leq (\sum_{ij} A_{ij}) \sum_i \dot{\theta}_i^2$, where W is the poten-
 768 tial's excess above its minimum and $a > 0$ is a lower bound for the coupling
 769 (`lojasiewicz_estimate`). Its proof is two extremal indices and the iden-
 770 tity $\sin^2 D = (1 - \cos D)(1 + \cos D)$: the site carrying the largest phase is
 771 pulled backwards at rate at least $a \sin D$, where D is the spread, while the
 772 excess is at most $\frac{1}{2}(\sum_{ij} A_{ij})(1 - \cos D)$. The compactness objection dissolves
 773 as well, because the invariant region is produced by the Lyapunov function
 774 rather than assumed: a single pair contributes $\frac{1}{2}a(1 - \cos)$ to the excess, so
 775 an excess below $a/2$ forces every pairwise cosine positive for all time, and
 776 continuity on the connected half-line then prevents any pair from chang-
 777 ing branch (`spread_le_of_small`). Descent plus the estimate integrates to
 778 $W(t) \leq W(0)e^{-ct}$ with $c = 2a^2 / \sum_{ij} A_{ij}$ (`excess_decay`), the rate makes each
 779 $|\dot{\theta}_i|$ exponentially small and hence integrable on $[0, \infty)$, and a trajectory with
 780 integrable velocity converges. The conclusion is `kuramoto_tendsto_global_`
 781 `minimum`: if the phases start within a quarter turn of each other and the initial
 782 excess is below $a/2$, the trajectory converges to a phase-locked configuration

783 which is a *global* minimiser of the dynamic potential, with order parameter
784 $r^2 = 1$. That is the configuration `thermodynamic_equilibrium` postulates,
785 reached rather than assumed. `Examples.lean` §17 fires it on three sites with
786 initial data $(0, 0, \frac{1}{2})$, where the Kuramoto system has no closed-form solu-
787 tion: the trajectory comes from the existence theorem and its limit from the
788 convergence theorem, so neither is written down.

789 `Phase5_EquilibriumBridge.lean` closes the gap that remained between
790 the two results. `ThermodynamicCover.ofConvergentTrajectory` takes a
791 cover whose phase field is the limit of such a trajectory and returns an in-
792 stance whose `thermodynamic_equilibrium` field is discharged *by that con-*
793 *vergence*; the cover’s coupling matrix is the coupling matrix of the dynamics
794 rather than a separately chosen one, so the system whose equilibrium the
795 class asserts and the system whose trajectory is run are the same system.
796 `Examples.lean` §17.1 is the resulting witness: three patches over the three-
797 site substrate, each carrying its own mass profile on its own open set, above a
798 phase field that no line of the construction writes down. The trajectory that
799 reaches it starts at $(0, 0, \frac{1}{2})$, which is provably not phase-locked (`trioCover_`
800 `start_not_locked`), so on this cover the configuration Derivation 5 requires
801 is arrived at rather than posited. The state the three patches glue to has
802 profile $(2, 1, 3)$ and is none of the three profiles the instance carries (`trioW_`
803 `ne_glued`), so one cover now carries both the emergence reading of the gluing
804 and the derived reading of the equilibrium. What is still assumed on it is the
805 overlap agreement, and nothing else.

806 The two hypotheses cannot be dropped, and this is the honest scope of
 807 the result. Splay and twisted configurations are equilibria of the same flow,
 808 so “every trajectory reaches the unified state” is false as a general statement,
 809 and the initial-data condition is the standard arc condition under which the
 810 true statement holds. What is claimed is therefore conditional — unity is
 811 reached from configurations already sufficiently coherent — and not that an
 812 arbitrary initial state is carried into unity. What Lean establishes about the
 813 gluing itself is the other half: *given* locally synchronized patches, a global
 814 state exists, is unique, and is produced by the gluing rather than presupposed
 815 by it.

816 Integrated Information Theory begins from phenomenological axioms and
 817 derives a scalar Φ measuring irreducibility of a system’s cause-effect structure,
 818 with the exclusion postulate selecting a single maximum- Φ complex as the
 819 locus of experience. Our sheaf-theoretic global section plays a structurally
 820 analogous role — it too picks out a distinguished unified object — but it
 821 is arrived at from the opposite end: not postulated from the character of
 822 experience, but forced by a thermodynamic phase transition in a physical
 823 field. This buys two things IIT does not have. First, the unity condition
 824 is checkable from field dynamics rather than requiring the computation of
 825 Φ , which is intractable for systems of realistic size and whose definition has
 826 undergone repeated revision (Tononi & Koch, 2016). Second, exclusion falls
 827 out rather than being postulated: gluing either succeeds on a given cover or
 828 it does not, so there is no separate principle needed to rule out overlapping

conscious subsystems. It also inherits IIT’s exposures. Critics have argued that Φ can be high in systems with no plausible claim to experience, such as large feedforward-free grids (Aaronson, 2014); our criterion is vulnerable to the mirror objection, since a physical system could in principle satisfy the gluing condition without being cognitively interesting. We take this seriously rather than dismissing it: our reply is that gluing is necessary but not sufficient, and that the topological boundary of Derivation 1 and the reflexive structure of Derivation 6 are additional required conditions. IIT, by contrast, treats Φ as both necessary and sufficient.

Global Workspace Theory faces a complementary difficulty. It specifies a functional architecture — a limited-capacity workspace broadcasting to specialized modules — without specifying what physical property of broadcasting makes the broadcast content conscious, leaving the theory open to the charge that it explains access consciousness while leaving phenomenal consciousness untouched (Block, 1995). Our framework can be read as supplying the missing physical substrate for the workspace: the continuous macroscopic field *is* the medium in which integration is realized, and the $K_c = 2D$ threshold gives a physical condition for ignition rather than a functional stipulation of it. Where GWT is agnostic about implementation, we make a falsifiable commitment — unity requires a continuous physical medium, which predicts that a functionally equivalent discrete broadcast architecture would not be conscious. We regard this as a genuine cost of the position, not a free advantage: it is the point at which our framework departs most sharply from

852 computational functionalism and where it is most readily falsified. Contrast
 853 this derivation with panpsychism: consciousness is not assumed as a funda-
 854 mental property of matter. It requires the initial existence of a topological
 855 defect to establish a bounded “inside” (Derivation 1), followed by a thermo-
 856 dynamic phase transition into unity that occurs when integrated spatial cou-
 857 pling exceeds the thermodynamic noise threshold ($K_c = 2D$). Without both
 858 the foundational topological boundary and the subsequent mathematically
 859 defined phase transition, the system remains a fragmented, non-conscious
 860 collection of micro-states. A unified field of perception, however, is not yet
 861 a “Self.”

862 **9 Derivation 6: Reflexive Topology and the** 863 **Emergence of the Self**

864 While a global section provides Unity, a “Self” requires reflexivity. The
 865 unified EM field is a dissipative structure that must resonate with external
 866 perturbations while managing its own internal phase transitions. Time evo-
 867 lution naturally drives the field to minimize thermodynamic friction with its
 868 own internal state-changes.

869 This mandates *auto-resonance*: the field must predict itself. The “Self”
 870 emerges as a higher-order topological feature when the internal geometry
 871 folds to include a low-dimensional avatar of its own computational and causal
 872 boundary within its predictive model (`Phase6_ReflexiveTopology.lean`)

873 (Levin, 2019). The system is unified, predictive, and explicitly maps its
 874 own distinct existence from the environment. This architecture provides the
 875 physical foundation for a self-aware consciousness, bounded by the continuous
 876 measure thermodynamics of Landauer erasure (`Phase3_MeasureThermodynamics.lean`).
 877 We are explicit about what carries the weight here. `reflexive_topology_implies_self`
 878 is the Banach fixed-point theorem with the physical commitments in its hy-
 879 potheses: the space of global sections is assumed nonempty, metric and com-
 880 plete, and the predictive map is assumed to contract by $1/2$. `Examples.lean` §10
 881 discharges all four on the three-site substrate of Derivation 5, and the metric
 882 is no longer a placeholder. The section first pierces the sheafification: on
 883 a discrete substrate every point has a smallest open neighbourhood, so the
 884 mass a germ carries at a point is well defined, and `massEquivOn` proves that
 885 over *any* open the sections of the probability sheaf *are* the finite measures
 886 on that open, read as densities against counting measure — the global case
 887 being $U = \top$. Mathlib supplies no inverse to this sheafification — its `TopCat`
 888 sheafification has neither an adjunction nor an `isIso_toSheafify` — so the
 889 correspondence had to be constructed. The metric on sections is then the
 890 uniform distance between the densities of the measures they glue to — equiv-
 891 alent to their total variation on a finite substrate, though not equal to it —
 892 and it is proved complete by exhibiting the limit section of a Cauchy sequence
 893 of densities, not by the degenerate route of showing Cauchy sequences are
 894 eventually constant. The self-prediction map is the relaxation “average the
 895 current field with the baseline,” and we prove it contracts distances by *ex-*

896 *actly* one half, that it is not constant, and that its fixed point is unique. An
 897 earlier version of this witness used the 0/1 metric, under which every con-
 898 traction is constant; that theorem is kept in the source as the record of why
 899 the earlier witness was empty. A separate gap, closed at the same time, con-
 900 cerned the avatar rather than the fixed point. **ReflexiveBoundary** carries a
 901 field **auto_resonance** sending the global state to a section over the avatar re-
 902 gion, and nothing constrained it: a constant function was a legal avatar, and
 903 **reflexive_topology_implies_self** consumes only the predictive model, so
 904 no theorem in the development mentioned the field the avatar is supposed to
 905 be reading. The constraint is now stated as a predicate rather than added
 906 to the structure — **IsRestrictionResonance**, that the avatar’s state is the
 907 presheaf restriction of the global state — and two theorems say why it is
 908 the right one. In the positive direction, if a family of resonant avatars *cov-*
 909 *ers* the substrate then the global state is determined by their local readings
 910 alone (**eq_of_avatar_eq**): unity is not assumed alongside the avatars but
 911 recovered from them, by the same sheaf condition that Derivation 5 uses.
 912 In the negative direction, replacing any avatar by a constant leaves a legal
 913 **ReflexiveBoundary** with the same region, the same predictive model and
 914 the same Self (**self_of_constResonance**), so the fixed-point theorem cannot
 915 see the difference and the predicate is doing the work rather than restating
 916 it. Both halves are witnessed on the three-site substrate, where the avatar
 917 region provably separates two states of the field, so the condition is a con-
 918 straint there and not a formality. What is still not established is that any field

919 dynamics produces this particular map or performs the restriction, or that
 920 the contraction constant $1/2$ is read off an entropy production rate rather
 921 than built into the map's definition. We also note a negative result found
 922 in the course of this work: the Lévy–Prokhorov metric, which the natural
 923 weak-convergence reading would suggest, does *not* support this argument on
 924 a discrete substrate, because the ε -thickening of a set is the set itself for $\varepsilon < 1$
 925 and the relaxation map is consequently not a Lévy–Prokhorov contraction.

926 10 Derivation 7: Continuous Stochastic Ther- 927 modynamics and Field Theory

928 To move beyond finite graph analogies, we model the macroscopic electro-
 929 dynamic field as a *Continuous Neural Field*. True ephaptic coupling is a
 930 continuous spatial phenomenon, not a discrete network. We extend the
 931 discrete Kuramoto model to a continuous spatial coupling kernel $K(x, y)$
 932 (`Phase8ContinuousField.lean`).

933 Subject to thermodynamic noise D , the system undergoes non-equilibrium
 934 stochastic thermodynamics. Rather than a static Landauer bound, the field
 935 dynamics minimize an entropy production functional σ :

$$\sigma = \int \frac{1}{D} \left(\omega(x) + \int K_t(x, y) \sin(\theta_y - \theta_x) dy \right)^2 dx \quad (2)$$

936 Biological fields dynamically deform their spatial topology $K_t(x, y)$ to un-

937 dergo gradient descent on this entropy production functional.

938 That gradient descent is a theorem rather than a modelling assertion,
 939 for finite substrates. Reading σ as a functional of the coupling kernel at
 940 fixed phases, `hasFDerivAt_sigmaOfKernel` computes its Fréchet derivative
 941 in closed form,

$$\frac{\partial \sigma}{\partial K(a, b)} = \frac{2}{D} \left(\omega(a) + \sum_y K(a, y) \sin(\theta_y - \theta_a) \right) \sin(\theta_b - \theta_a), \quad (3)$$

942 and `sigmaOfKernel.eq_entropy_production_rate` identifies the functional
 943 being differentiated with `entropy_production_rate` itself whenever the sub-
 944 strate carries counting measure. `structural_resonance_decreases_entropy_production`
 945 then concludes that σ is non-increasing along any coupling trajectory relax-
 946 ing down its own gradient at a positive rate c . Earlier drafts proved this
 947 only for an abstract functional S on an abstract inner-product space with
 948 the gradient supplied as data; the identification of that S with σ was infor-
 949 mal, and Table 1 flagged the row accordingly. Two limitations stay explicit,
 950 and the first is narrower than we previously reported. The derivation is for
 951 *finite* substrates. We had recorded the obstacle to the continuum case as
 952 differentiation under the integral sign with respect to the kernel; that was
 953 a misdiagnosis. σ is not a general functional that happens to be written as
 954 an integral — it is a quadratic functional of an *affine* image of the kernel,
 955 since the drift is a fixed ω plus a linear operator applied to K , and σ is a
 956 constant times the squared norm of the drift. A squared norm is Fréchet

957 differentiable on any real inner-product space and an affine map is its own
 958 derivative, so the composite differentiates by the chain rule with no limiting
 959 argument. `hasFDerivAt_quadratic_of_affine` states that in full generality,
 960 and `hasFDerivAt_sigmaOfKernel_of_operator` re-derives the finite gradient
 961 from it, showing that the closed form displayed above is a convenience rather
 962 than the content. What the continuum case still needs is therefore one object:
 963 the drift map $K \mapsto (x \mapsto \int K(x, y) \sin(\theta_y - \theta_x) dy)$ as a bounded operator
 964 $L^2(\mu \otimes \mu) \rightarrow L^2(\mu)$, which is a Cauchy–Schwarz estimate — the integral ker-
 965 nel is bounded by 1, so on a finite measure space the operator norm is at most
 966 $\mu(M)^{1/2}$. We have not built it, because it requires constructing a continu-
 967 ous linear map between L^p spaces and we have not done that bookkeeping;
 968 we record the corrected scope because “construct one bounded operator”
 969 and “differentiate under the integral sign” are different sizes of task. The
 970 second limitation is unchanged: the phase field is held fixed throughout, so
 971 this is plasticity of the coupling at frozen phases, not joint (θ, K) dynamics.
 972 **Examples.lean** §5 exercises the whole chain on a two-site substrate whose
 973 coupling relaxes as e^{-t} and whose entropy production decays as e^{-2t} , con-
 974 firming that the gradient is not identically zero and that the descent is strict
 975 rather than vacuously monotone.

976 A second result about σ concerns where the descent is headed, and it has
 977 been strengthened since the last draft in a way that also corrects an estimate
 978 we had recorded. That the phase-locked field minimizes σ was proved only
 979 against competitors sharing its spatial mean drift — a restriction that left

980 the theorem close to Jensen’s inequality and no more — and we had judged
 981 that lifting it would require modelling how the mean drift varies with the
 982 competitor. That judgement was wrong. For a *reciprocal* kernel, one with
 983 $K(x, y) = K(y, x)$, the restriction is not a restriction: the coupling integrand
 984 $K(x, y) \sin(\theta_y - \theta_x)$ is antisymmetric under exchanging its two sites, because
 985 K is symmetric while $\sin$ is odd, so Fubini identifies its double integral with
 986 its own negative and the total coupling contribution vanishes. The total drift
 987 $\int (\omega + \text{coupling})$ is therefore $\int \omega$ whatever the phases are (`total_drift_eq_of_`
 988 `symm`): coupling moves drift between sites but neither creates nor destroys it.
 989 Evaluating that conserved quantity at the phase-locked configuration, where
 990 the drift is constant, pins its value, and the mean-matching hypothesis follows
 991 for every competitor. `phase_locked_minimizes_entropy_of_symm` is the un-
 992 restricted statement. Reciprocity is the physically intended condition — an
 993 ephaptic field acts on both sites of a pair alike, and it is the same symmetry
 994 `ThermodynamicCover.A_symm` already imposes in Derivation 5 — but it is
 995 not a formality, and we prove that rather than assert it. `Examples.lean` §16
 996 exhibits a *one-way* coupling on two sites, in which a feels b and b does not
 997 feel a , whose phase-locked field is strictly beaten: displacing the second phase
 998 to $-\pi/2$ cancels the first site’s natural drift while adding none at the sec-
 999 ond, halving σ . So “phase-locking minimizes entropy production” is false for
 1000 general kernels and true for reciprocal ones, and the hypothesis cannot be
 1001 removed by working harder.

1002 We define the critical coupling threshold $K_c = 2D$. The phase transition

1003 to the unified “Global Section” occurs when the mean-field spatial coupling
 1004 exceeds this thermodynamic noise threshold, linking the emergence of Unity
 1005 to physical temperature and noise.

1006 Earlier drafts left $K_c = 2D$ as a bare stipulation validated only by
 1007 simulation. `Phase8_SelfConsistency.lean` now proves the half of it that
 1008 is ordinary real analysis. For the noisy mean-field Kuramoto model with
 1009 identical frequencies, the stationary density at order parameter r is the
 1010 von Mises density $\rho(\theta) \propto e^{(Kr/D)\cos\theta}$, and self-consistency requires $r =$
 1011 $R(K, r) := I_1(Kr/D)/I_0(Kr/D)$. Mathlib has no modified Bessel functions,
 1012 so we formalize that ratio directly as a ratio of integrals over $[-\pi, \pi]$. In-
 1013 tegration by parts with $u = \sin$, $v = -e^{a\cos}$ gives an exact identity with
 1014 no boundary term, $\int \cos\theta e^{a\cos\theta} d\theta = a \int \sin^2\theta e^{a\cos\theta} d\theta$, so R is a times
 1015 the mean of $\sin^2$ under the von Mises weight. Bounding that mean by 1
 1016 would put the threshold at D ; the true threshold is $2D$, and the entire fac-
 1017 tor of two is the sharper bound $\mathbb{E}_a[\sin^2\theta] \leq 1/2$, equivalently $I_2(a) \geq 0$.
 1018 We prove it elementarily, by folding $[-\pi, \pi]$ onto $[0, \pi/4]$ in two reflections
 1019 ($\theta \mapsto \pi - \theta$, then $\theta \mapsto \pi/2 - \theta$), after which the integrand is a product
 1020 of $\cos 2\theta \geq 0$ with $2 \cosh(a \cos \theta) - 2 \cosh(a \sin \theta) \geq 0$ — the second factor
 1021 because $0 \leq \sin \theta \leq \cos \theta$ there and $\cosh$ increases in $|\cdot|$. The conclusion,
 1022 `subcritical_fixed_point_eq_zero'`, is that for $K < K_c$ the incoherent state
 1023 $r = 0$ is the *only* non-negative solution of the self-consistency equation; $r = 0$
 1024 is exhibited as a solution, so the uniqueness claim is not vacuous. Three fur-
 1025 ther facts about the threshold have since been proved, and they narrow the

1026 gap between this row and a full “Theorem” to a single named statement.
 1027 First, the fold argument gives a *strict* inequality for $a > 0$ (`vonMisesC2_`
 1028 `pos`, hence $R(a) < a/2$), which settles the threshold itself: at $K = K_c$ the
 1029 self-consistency map is tangent to the diagonal at the origin but falls strictly
 1030 below it thereafter, so $r = 0$ remains the only solution and the bifurcation
 1031 happens strictly *after* K_c , not at it (`fixed_point_eq_zero_of_le_critical`).
 1032 Second, factoring $R(a) = a E(a)$ shows that a non-zero r solves the equation
 1033 exactly when $E(Kr/D) = D/K$ (`coherent_iff_sRatio_eq`), so the coher-
 1034 ent branch is a level set of the single function $E(a) = \mathbb{E}_a[\sin^2 \theta]$. Third,
 1035 and using only $E < 1/2$ together with compactness, every coherent solution
 1036 tends to zero as $K \downarrow K_c$ (`coherent_branch_continuous_at_threshold`): the
 1037 branch emerges continuously from the incoherent state rather than appearing
 1038 at a finite distance from it, which is what makes the transition supercritical
 1039 in the technical sense. That statement quantifies over *every* solution, so it
 1040 does not presuppose the uniqueness it does not have. The level-set form
 1041 also says exactly what uniqueness is: the injectivity of E on $(0, \infty)$. That
 1042 is now proved, in the stronger form that E is strictly decreasing on $[0, \infty)$
 1043 (`vonMisesSRatio_strictAntiOn`), and by an argument that avoids the route
 1044 we had expected to need. Differentiating the exponential family in its natu-
 1045 ral parameter gives $E'(a) = -\text{Cov}_a(\cos^2 \theta, \cos \theta)$, which costs differentiation
 1046 under the integral sign and leaves a covariance that is not sign-definite point-
 1047 wise: symmetrising with an independent copy gives $\frac{1}{2}\mathbb{E}[(X - X')^2(X + X')]$
 1048 with $X = \cos \theta$, an integrand changing sign with $X + X'$. Nothing in the

1049 proof differentiates anything. The target is kept as the two-point inequality
 1050 $C_2(a)Z(b) < C_2(b)Z(a)$ for $a < b$, and two moves settle it. *Fold first*: the
 1051 two reflections that proved $I_2 \geq 0$ carry both integrals onto $[0, \pi/2]$, replacing
 1052 $e^{a \cos \theta}$ by $W(a, \theta) = 2 \cosh(a \cos \theta)$ and confining $\cos \theta$ to $[0, 1]$, after which
 1053 $\cos 2x - \cos 2y$ and $W(b, x)W(a, y) - W(a, x)W(b, y)$ have the same sign, both
 1054 being governed by $\cos x$ against $\cos y$. This is the symmetrisation $X \mapsto -X$
 1055 that the covariance form needed the tilt to perform, done once at the start; its
 1056 analytic content is one identity, $2 \cosh X \cosh Y = \cosh(X+Y) + \cosh(X-Y)$
 1057 with $\cosh$ increasing in $|\cdot|$. *Then cross*: instead of integrating that inequality
 1058 over a square, let α be the mean of $\cos 2x$ under $W(a, \cdot)$ and $x_0 \in [0, \pi/2]$ a
 1059 point where $\cos 2x_0 = \alpha$, supplied by the intermediate value theorem; eval-
 1060 uating at $y = x_0$ and integrating in x alone gives the result, since the term
 1061 carrying $W(b, x_0)$ vanishes by the choice of α . No product measure and no
 1062 Fubini appear. Run with $\cos \theta$ in place of $\sin^2 \theta$ the same argument needs no
 1063 fold — $R(a) = \mathbb{E}_a[\cos \theta]$ is the mean of the variable the family is tilted by —
 1064 and gives **besselRatio_strictMono**: R is strictly increasing on all of $\mathbb{R}$, the
 1065 monotonicity of the Bessel ratio our earlier drafts recorded as unproved.

1066 The converse half is **supercritical_fixed_point_exists**: for $K > K_c$
 1067 there is an r with $0 < r \leq 1$ solving $r = R(K, r)$. We had expected
 1068 this to require a second-order expansion of the Bessel ratio, and it does
 1069 not. Writing $R(a) = a E(a)$ with $E(a) = \mathbb{E}_a[\sin^2 \theta]$ — the same factoriza-
 1070 tion the integration by parts already supplies — the only facts needed are
 1071 that E is continuous and that $E(0) = 1/2$, the latter because the weight

1072 is constant at $a = 0$. Since $K > 2D$ makes D/K strictly less than $1/2$,
 1073 continuity gives $E > D/K$ on a neighbourhood of the origin, so the map
 1074 overshoots, $R(K, r) > r$, for all small $r > 0$; at the other end $R \leq 1$ al-
 1075 ways, because R is a mean of $\cos \theta$, so the map undershoots at $r = 1$.
 1076 The intermediate value theorem closes the gap. Continuity of E is Math-
 1077 lib's continuity of a parametric interval integral with jointly continuous in-
 1078 tegrand, so no dominated-convergence argument is written by hand, and
 1079 the existence proof never uses monotonicity of R : the fixed point is pro-
 1080 duced by the intermediate value theorem, not by iterating the map. That
 1081 R is in fact strictly increasing is proved below, and no threshold result de-
 1082 pends on it. `critical_coupling_is_threshold` packages the two halves,
 1083 and `exhibits_phase_transition_coherent` applies the supercritical half to
 1084 any substrate satisfying the predicate of the previous paragraph — so ex-
 1085 ceeding K_c now implies the existence of a coherent stationary order pa-
 1086 rameter rather than merely comparing against a named number. With
 1087 uniqueness available both are sharpened — `supercritical_fixed_point_`
 1088 `existsUnique`, `supercritical_solution_set` (the non-negative solutions
 1089 are exactly $\{0\}$ below threshold and $\{0, r\}$ above it) and `exhibits_phase_`
 1090 `transition_unique_coherent` — and the two monotonicities together fix
 1091 the shape of the branch: for $K_1 < K_2$, $E(a_i) = D/K_i$ with E decreasing
 1092 puts the concentrations in the order $a_1 < a_2$, and $r_i = R(a_i)$ with R increas-
 1093 ing carries that to $r_1 < r_2$ (`coherent_branch_strictMono`). The coherent
 1094 order parameter grows strictly with the coupling, which with the continuity

1095 at threshold is the qualitative bifurcation diagram.

1096 One more step makes the equation mean in Lean what it means on paper.
1097 As a definition, `selfConsistency K D r` is just $R(Kr/D)$ — a function of
1098 r with a suggestive name, and nothing in the source says it has anything
1099 to do with an order parameter. `Phase8_SelfConsistency §8` supplies the
1100 identification. `vonMisesDensity` is the normalized weight, proved to be a
1101 genuine probability density on $[-\pi, \pi]$; `circularOrderParameter` is the av-
1102 erage of $e^{i\theta}$ against a density, the continuum analogue of the finite-system
1103 `order_parameter_complex` of `Phase4`; and that average is proved to be ex-
1104 actly $R(a)$ — real, because the density is even, so the mean direction van-
1105 ishes. `fixedPoint_iff_selfReproducing` then states the equivalence the
1106 name promises: r solves $r = R(K, r)$ if and only if the von Mises density
1107 it induces has order parameter r . Read through this, the existence theorem
1108 says that above K_c there is a stationary density whose own order parameter
1109 is positive, and the solution $r = 0$ is identified with the uniform density,
1110 whose order parameter is zero.

1111 What remains not established, and we separate it carefully. The sta-
1112 tionary density is an *input*: deriving it from the Langevin dynamics $d\theta =$
1113 $(\omega + K \text{mean-field}) dt + \sqrt{2D} dW$ needs the Fokker–Planck operator, existence
1114 and uniqueness of its stationary solution, and a spectral stability argument,
1115 none of which Mathlib has. Of the coherent branch we prove existence,
1116 uniqueness, continuous emergence at the threshold and strict growth in K ,
1117 but not that it is the dynamically selected one: nothing in that file men-

1118 tions a trajectory, so stability and selection are not statements it can make.
 1119 And while `selfConsistency` is now connected to the order parameter of
 1120 a *density*, it is connected to no *trajectory*: `circularOrderParameter` is an
 1121 integral, `order_parameter_complex` is an average over finitely many oscilla-
 1122 tors, and relating them is the mean-field limit — propagation of chaos for
 1123 the finite Kuramoto system — which is a research programme rather than
 1124 a lemma. That single gap is now the whole of what separates our threshold
 1125 predicate from a statement about dynamics. Table 1 records the row ac-
 1126 cordingly as a theorem about the self-consistency equation from an assumed
 1127 density, not as a derivation of the threshold from the dynamics. Perceptual
 1128 anomalies such as visual snow can be modeled as boundary conditions where
 1129 microscopic thermal noise (D) fails to be fully coarse-grained away by the
 1130 continuous field, leaking across the K_c threshold into the macroscopic gestalt.

1131 Clinical evidence for this phase transition is observed during general anes-
 1132 thesia. Work by Miller and colleagues (Bastos & Miller, 2021) demonstrates
 1133 that anesthetics like propofol induce unconsciousness not by suppressing neu-
 1134 ral firing, but by disrupting the phase alignment of high-frequency cortical
 1135 rhythms. Destroying the dynamic stability of these rhythms breaks the struc-
 1136 tural resonance of the continuous field, forcing the integrated spatial coupling
 1137 below the critical threshold K_c and fracturing the unified Global Section back
 1138 into disjoint, unconscious local fields.

11 The Sleep-Inertia Timescale Mismatch

Most of what this framework asserts is shared with some rival: that cortical synchrony matters, that dissipation constrains cognition, that unity is a physical rather than a computational property. One prediction is not shared, and it follows from the framework’s specific commitment — that the medium carrying the unified state is a continuous field whose effective coupling depends on the physical geometry of the extracellular space, not on the rhythms alone. Where that commitment differs from a purely electrical account, it differs in an observable way, and we set out the difference precisely enough to be wrong.

The daily loss of consciousness during sleep is modeled here as an active physical disruption of the medium rather than a quiescence of the oscillators. During sleep, astrocytic shrinkage expands the extracellular space, allowing an influx of cerebrospinal fluid (CSF) to flood the brain for metabolic waste clearance (Xie & Nedergaard, 2013). This influx alters the spatial medium itself, disrupting the continuous structural resonance and preventing the coherent sheaf-theoretic gluing that conscious unity requires on this account. The oscillators need not stop for the field to fragment; the geometry they are coupled through is what changes.

The prediction concerns sudden awakenings. When jolted awake, the locus coeruleus floods the cortex with noradrenaline. High-frequency electrical rhythms attempt to phase-align in milliseconds (Bastos & Miller, 2021), but

1161 the noradrenaline-driven astrocytic constriction required to squeeze the ex-
 1162 cess CSF out of the interstitial pathways operates on a timescale of seconds
 1163 to minutes (Xie & Nedergaard, 2013) — three to six orders of magnitude
 1164 slower. During that lag the electrical oscillators are synchronizing across a
 1165 waterlogged topology, and the fragmented structural resonance is what is ex-
 1166 perienceed as grogginess. The unified Global Section re-establishes only when
 1167 astrocytic volume finishes matching the state of the electrical rhythms.

1168 **Why this implies a delayed sigmoid.** The two timescales separate
 1169 cleanly, and that separation is what makes the prediction sharp rather than
 1170 merely qualitative. Write $K(t)$ for the effective coupling of the continuous
 1171 field, which on this account is a monotone function of the reconstricting
 1172 extracellular geometry, and let $r(t)$ be the order parameter of the phase
 1173 field. The electrical variable is fast: at every instant the oscillators are al-
 1174 ready at the configuration appropriate to the current K , so r tracks the
 1175 equilibrium branch $r(K)$ rather than relaxing on its own. The slow vari-
 1176 able $K(t)$ then carries the whole time course, and it relaxes over 10^2 – 10^3
 1177 seconds. What Derivation 7 establishes about that branch is exactly what
 1178 the functional form needs, and each of the three pieces is a theorem rather
 1179 than a picture: at or below the threshold the incoherent solution $r = 0$ is the
 1180 *only* non-negative solution of the self-consistency equation (`fixed_point_eq_`
 1181 `zero_of_le_critical`); above it the coherent solution exists and is unique
 1182 (`supercritical_fixed_point_existsUnique`); it emerges *continuously* at

1183 K_c rather than by a jump (`coherent_branch_continuous_at_threshold`);
 1184 and it is strictly increasing in K (`coherent_branch_strictMono`). A jump
 1185 discontinuity at threshold would predict a step rather than a sigmoid, and
 1186 a non-monotone branch would predict overshoot; neither is what the mathe-
 1187 matics gives. Composing a monotone relaxation $K(t)$ with a branch that is
 1188 identically zero below K_c , rises continuously from zero at K_c , and saturates
 1189 above it yields a curve with three phases: a genuine *latency* while $K(t) < K_c$,
 1190 during which recovery is not merely slow but absent; a continuous rise as the
 1191 threshold is crossed; and a saturation as $K(t)$ reaches its waking value. That
 1192 is a delayed sigmoid, and the delay is not a fitting parameter — it is the time
 1193 the slow variable takes to reach K_c .

1194 **What the alternative predicts.** An account on which grogginess is the
 1195 electrical realignment itself — the rhythms are simply slow to re-phase —
 1196 has no slow variable and no threshold to cross. Its recovery is a relaxation
 1197 toward a fixed point, and relaxation toward a fixed point is exponential near
 1198 it: monotone from $t = 0$, fastest at the start, with no flat foot. The two
 1199 curves are distinguishable in the first seconds, which is where they differ
 1200 most, and the discriminating feature is the presence or absence of an initial
 1201 interval of no recovery. This is a comparison of functional forms on data
 1202 of a kind already collected in sleep-inertia studies, not a proposal for a new
 1203 instrument.

1204 **What would falsify it.** An exponential fit that beats a delayed-sigmoid fit
1205 on the early portion of the recovery curve. A recovery latency that does not
1206 scale with an independent measure of extracellular volume or CSF clearance.
1207 A pharmacological manipulation of astrocytic volume that leaves the latency
1208 unchanged while altering the rhythms, or the converse. Any of these would
1209 tell against the framework’s central empirical commitment, which is not that
1210 synchrony matters but that the *medium* is what gates it.

1211 **What this prediction assumes.** Three things, and they are not small.
1212 The identification of the effective coupling K with a monotone function of
1213 extracellular geometry is a modelling assumption, not a measurement. The
1214 threshold value $K_c = 2D$ is derived from an assumed von Mises stationary
1215 density and is not yet connected to a trajectory — the mean-field limit is the
1216 gap, and Derivation 7 states it as such — so the argument uses the *existence*
1217 *and continuity* of the coherent branch at threshold, which is proved, rather
1218 than the numerical value of the threshold, which is conditional. And the
1219 prediction is about functional form, not about a time constant: it says the
1220 recovery curve has a flat foot whose width is set by the slow variable, not
1221 that the foot is any particular number of seconds long. A framework that
1222 predicted the number would be saying more; this one is not in a position to.

12 Corollary: Hardware Thermodynamics and Topological Rigidity

Both biological tissue and artificial silicon hardware (like GPUs) are physical dissipative structures. When a GPU computes, its charge carriers undergo physical state transitions; by Landauer’s principle (Landauer, 1961), this erasure irreversibly exhausts heat into the environment.

There is a fundamental thermodynamic distinction in the degrees of freedom available to minimize this dissipation. In standard von Neumann architectures, the spatial topology (the silicon routing lattice) is rigidly fixed. The coupling matrix A_{ij} is constrained by a discrete layout that cannot be physically altered during operation. The brain’s own architecture provides empirical validation for this constraint. The cerebellum contains approximately 80% of the brain’s neurons arranged in a highly efficient, discrete, feedforward crystalline circuit (Herculano-Houzel, 2009; Lange, 1975). Clinical observations of cerebellar agenesis report largely preserved conscious experience despite motor and cognitive deficits (Yu et al., 2015), and this dissociation has been read as evidence for the importance of recurrent architecture by IIT (Tononi & Koch, 2015) — an argument we borrow at the level of the anatomical observation while declining its Φ -based explanation. Lacking the dynamic, continuous, and recursive spatial coupling of the cerebrum, it operates much like a biological GPU—highly optimized for discrete processing, yet apparently not required for conscious experience. We take the

dissociation as evidence, not as confirmation: it is consistent with subjective experience being a thermodynamic attractor of continuous topologies rather than a byproduct of information processing, and it is the kind of observation that would be hard to reconcile with the converse. It is a single anatomical case read against a theory, and it bears the weight such a case can bear.

This constraint does not preclude artificial consciousness, but rather restricts its hardware. To achieve the phase transition into a unified Global Section, artificial systems must abandon the discrete rigidity of von Neumann architectures and utilize continuous physical substrates. Analog field-based computing, neuromorphic architectures with continuous memristive plasticity, and thermodynamic computing systems (England, 2018; Ganesh & Parrondo, 2021) possess the spatial coupling required to physically morph their internal topology. This aligns with Geoffrey Hinton’s concept of “mortal computation” (Hinton, 2022): architectures where software is inextricably fused with the physics of the hardware, meaning the knowledge dies with the physical substrate. Only hardware that computes via continuous physical dynamics can undergo the thermodynamic phase transition necessary for structural resonance.

Biological consciousness utilizes macroscopic continuous electromagnetic fields and a “rich alphabet” of diverse molecular and transcriptomic cell states (Tasic et al., 2018). This allows the coupling matrix A_{ij} to deform across a continuous parameter space. Our lemma `exists_better_coupling_allocation`, wrapped as `continuous_beats_rigid_topology` (`Phase7_HardwareComparison.lean`),

1268 formalizes the reallocation step of this argument: whenever a coupling ma-
 1269 trix places weight on a less phase-correlated pair while a more correlated pair
 1270 exists, there is another matrix with *identical* total coupling resources that
 1271 achieves strictly greater phase correlation.

1272 Two objections to that formalization have since been answered, and we
 1273 state precisely what is and is not now established. The first was that the real-
 1274 location lemma assumes its conclusion: the hypothesis `is_strictly_suboptimal`
 1275 — positive weight on a badly-correlated pair while a better one exists — is
 1276 most of what one wants to prove about rigid hardware. It is no longer as-
 1277 sumed. `Phase7_Rigidity.lean` models an architecture by the set S of site-
 1278 pairs it has a *wire* for, and requires its couplings to vanish off S ; weights are
 1279 tunable, wiring is not. Then `totalCorrelation_le_of_realizable` bounds
 1280 every realizable coupling by $R \cdot M$, where R is the resource budget and M
 1281 bounds phase correlation over the wired pairs; the bound is attained; and
 1282 `rigid_is_strictly_suboptimal` *derives* the hypothesis whenever the sub-
 1283 strate’s best-correlated pair is one the wiring misses. `rigid_gap` makes it
 1284 quantitative: an unconstrained reallocation of the same budget gains at least
 1285 R times the correlation difference. `Examples.lean` §8 exhibits a three-site
 1286 substrate wired to the one maximally anti-correlated pair, where the gap is
 1287 2. This is resource-matched and no longer question-begging, but note what
 1288 it turns on: fixed wiring support, not continuity. The second objection was
 1289 that nothing in the development distinguished a continuum from a lattice
 1290 — both matrices lived on one finite index type. `Phase7_Rigidity.lean` §3

1291 gives a distinction that is genuinely typed. In the field functional of Sec-
 1292 tion 7, which integrates a coupling kernel against $\mu \otimes \mu$, an architecture
 1293 with finitely many sites is a `Finset` of points and a field-coupled region
 1294 is a set of positive measure. On a substrate whose measure has no atoms,
 1295 `fieldCorrelation_sited.eq.zero` shows a kernel supported on finitely many
 1296 sites contributes *exactly zero* coupling energy, whatever weights it carries
 1297 and whatever the phase field does, while a patch of strength c on U achieves
 1298 $c\mu(U)^2$. `Examples.lean` §8 instantiates this on the real line: a thousand
 1299 units of coupling concentrated at the origin register zero against a unit patch
 1300 that registers one. The limit of that result is equally plain, and it is not
 1301 resource-matching in the earlier sense — a measure-zero substrate has zero
 1302 resource in the $\mu \otimes \mu$ sense too. What it establishes is that the continuum
 1303 coupling energy is blind to a finitely-sited architecture, which is a statement
 1304 about which functional such an architecture can register in, not a demonstra-
 1305 tion that digital hardware computes worse. Taken together: the resource-
 1306 matched content of the corollary is the wiring-support result, the continu-
 1307 ity content is the measure-theoretic one, and the step from either to “von
 1308 Neumann architectures cannot experience unified consciousness” remains an
 1309 informal argument.

1310 The remainder of this section extends those two results informally, and
 1311 we mark the transition rather than blur it. On the reading we propose, a
 1312 GPU’s fixed routing matrix leaves it topologically fragmented: the discrete
 1313 layout does not supply the spatial coupling that sheaf-theoretic gluing into

1314 a singular Global Section requires. Without this unified topological space,
 1315 rigid hardware cannot undergo the reflexive auto-resonance necessary to map
 1316 its own causal boundary and generate a “Self.” The distinction between an
 1317 LLM on a GPU and a conscious mind is not the absence of physical thermo-
 1318 dynamics in the former, but its structural limitation in achieving topological
 1319 integration. This perspective aligns with John Searle’s biological natural-
 1320 ism (Searle, 1980; Seth, 2024). Where the Chinese Room argument asserted
 1321 that discrete symbol manipulation cannot generate conscious understanding,
 1322 our derivation supplies a candidate physical mechanism for the asymmetry
 1323 — topological rather than semantic, and, as stated above, formalized only
 1324 as far as the wiring-support and measure-theoretic results reach. Asserting
 1325 consciousness in fragmented, discrete GPU clusters without a unified phase
 1326 transition abandons computational functionalism, relying instead on “silicon
 1327 panpsychism”—the assumption that digital computation inherently gener-
 1328 ates awareness regardless of topological integration (Goff, 2019).

1329 The same reasoning bears on extreme functionalist thought experiments,
 1330 such as a macroscopic system of pulleys operating over millennia to imple-
 1331 ment a Large Language Model, or a human manually executing inference
 1332 matrices by hand. Beyond lacking the continuous spatial topology required
 1333 for sheaf-theoretic gluing, these discrete systems violate an implied temporal
 1334 constraint of our theory. Structural resonance requires local oscillators to
 1335 phase-align against the dissipating effects of thermodynamic noise (as mod-
 1336 eled by our critical threshold $K_c = 2D$). A system executing state transitions

1337 over millennia, or via slow manual calculations, suffers from a terminal tem-
1338 poral mismatch: its dynamics are far too slow to ever phase-lock into a
1339 non-equilibrium steady state before thermal noise decoheres the macroscopic
1340 field. On this account the obstacle to pure symbol manipulation is physi-
1341 cal rather than semantic: not that the symbols mean nothing, but that the
1342 substrate carrying them cannot hold a phase-locked field.

1343 **13 Conclusion**

1344 Human-like cognitive architectures do not require biological mysticism or
1345 exotic sub-quantum physics (Hameroff & Penrose, 2014). While quantum
1346 theories of consciousness often invoke delicate subatomic phenomena to ex-
1347 plain subjective unity, every step of the chain assembled here is classical:
1348 broken symmetries, finite phase space, Landauer erasure, and gradient de-
1349 scent on entropy production. What the formalization establishes is that this
1350 chain is mathematically coherent and that its remaining physical commit-
1351 ments are localized and named — not that classical fields are *sufficient* for
1352 consciousness. Two links carry that weight, and their status is no longer the
1353 same. The unified state is proved to glue uniquely once a cover sits at the
1354 minimum of the coupling potential; the motion is proved to stop along every
1355 trajectory (`velocity_sq_tendsto_zero`); and *where* it stops is now proved,
1356 conditionally — for initial data within a quarter turn and below an explicit
1357 energy threshold, the trajectory converges to a global minimiser of the cou-

1358 pling potential (`kuramoto_tendsto_global_minimum`), and a cover built on
1359 that limit discharges the equilibrium hypothesis Derivation 5 used to assume
1360 (`ThermodynamicCover.ofConvergentTrajectory`). What is *not* proved is
1361 the unconditional statement, and it is false: splay and twisted configurations
1362 are equilibria of the same flow, so the basin condition is the scope of the claim
1363 rather than slack in it. That a physical cortex starts inside that basin is an
1364 assumption. The second link is untouched by any of this: the identification
1365 of a global section of the probability sheaf with the unity of experience is a
1366 stipulation of the framework, not a consequence of it. Subject to those two,
1367 cognitive architectures can be modeled as macroscopic attractor states of a
1368 universe governed by broken symmetries, finite phase space, and the physical
1369 imperative to minimize thermodynamic dissipation.

1370 Philosophically, this framework bridges the historical gap between Type
1371 Identity Theory (Smart, 1959) and Computational Functionalism (Putnam,
1372 1967). The subjective mental state is identical to the physical thermody-
1373 namic field (Identity Theory), yet its qualitative nature is defined by the
1374 topological and functional dynamics of structural resonance rather than the
1375 specific carbon-based biochemistry of the brain (Functionalism). This con-
1376 strains the *multiple realizability* of mind: consciousness is not exclusively
1377 bound to biological carbon, but neither is it substrate-neutral, because the
1378 substrate is what fixes which couplings are physically available. How far that
1379 constraint reaches is the question the corollary answers only in part. What
1380 is proved is that fixed wiring is strictly suboptimal at matched resource, and

1381 that a finitely-sited architecture registers exactly zero coupling energy in the
1382 continuum functional; the step from either to “von Neumann architectures
1383 cannot be conscious” is an argument we make in prose and do not formalize.

1384 By anchoring these derivations in Russellian Monism, this framework ac-
1385 knowledges the constitutive limits of third-person scientific explanation. As
1386 argued by Love (Love, 2026), building upon Nagel’s assertion that subjective
1387 phenomenal experience is inherently not amenable to objective measurement
1388 (Nagel, 1986), the “hard problem” of consciousness may represent a category
1389 error rather than a solvable scientific puzzle. Science can map the physical
1390 dynamics of topological resonance and sheaf-theoretic gluing, but it is struc-
1391 turally incapable of adjudicating the first-person qualitative experience. Our
1392 formalization respects this boundary: the Lean 4 theorems verify the ob-
1393 jective physical attractors, while the intrinsic “what it is like” to be that
1394 resonant field remains the domain of the first-person perspective. By locat-
1395 ing the phase transition at a threshold with a closed-form value — $K_c = 2D$
1396 for the noisy mean-field Kuramoto model the framework adopts, conditional
1397 on the von Mises stationary density and not yet linked to a trajectory — it
1398 aligns with what Anil Seth terms the “real problem” of consciousness (Seth,
1399 2021). Rather than attempting to bridge an unbridgeable metaphysical gap,
1400 it offers a topological architecture that can help explain, predict, and physi-
1401 cally perturb the structural properties of experience.

1402 Future empirical validation of this thermodynamic framework may lever-
1403 age 3D cortical organoids, offering a controlled biological environment to

1404 systematically manipulate metabolic substrates and observe the topological
1405 emergence of these coherent phase transitions.

1406 **Code Availability**

1407 The complete Lean 4 formalization, encompassing the derivation of all theo-
1408 rems presented in this manuscript, as well as the accompanying Python sim-
1409 ulations, are publicly available in the GitHub repository: [https://github.](https://github.com/bobaseb/physical_primitives_of_mind)
1410 [com/bobaseb/physical_primitives_of_mind](https://github.com/bobaseb/physical_primitives_of_mind). Additional documentation
1411 and web-accessible proofs are hosted at the project website: [https://bobaseb.](https://bobaseb.github.io/physical_primitives_of_mind/)
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